

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Technology has changed the way law firms manage their everyday work. Many firms now use computers, cloud storage, email, and online communication to create, store, and share legal documents. Lawyers can also communicate with clients, access case files, and work remotely using digital tools. These technologies make work faster, more organized, and more efficient. However, as law firms become more dependent on technology, they also face greater risks from cyber threats such as phishing, ransomware, malware, hacking, and data breaches that can compromise sensitive information (Verizon, 2025). Those attacks can cause important legal documents to be lost and may expose private client information without permission.

Law firms handle a large amount of sensitive and confidential data, including personal information, financial records, contracts, court documents, and legal strategies. Clients trust law firms to keep this information safe and private. If confidential data is stolen, altered, or leaked, it may harm clients, damage ongoing legal cases, and affect the reputation of the law firm. It may also result in financial losses and legal liabilities. Because of these risks, protecting digital information has become one of the most important responsibilities of every law firm.

Cybersecurity is the practice of protecting computers, networks, software, and digital information from unauthorized access, cyberattacks, and data theft. According to the National Institute of Standards and Technology (NIST, 2024), cybersecurity helps organizations protect

the confidentiality, integrity, and availability of information. Confidentiality ensures that only authorized individuals can access sensitive data. Integrity protects information from being changed or damaged without permission, while availability ensures that information is accessible whenever it is needed. By implementing effective cybersecurity measures, organizations can reduce risks, protect valuable information, and continue their operations without major disruptions (National Institute of Standards and Technology [NIST], 2024).

Aside from protecting their computer systems, law firms must also comply with laws and regulations related to data privacy and information security. In the Philippines, the Data Privacy Act of 2012 (Republic Act No. 10173) requires organizations to protect personal information and apply reasonable organizational, physical, and technical security measures (Republic of the Philippines, 2012). The law also promotes transparency, accountability, and responsible handling of personal data. According to the National Privacy Commission (2022), organizations that fail to comply with these requirements may face administrative penalties, legal sanctions, financial losses, and damage to their professional reputation. Since law firms regularly collect and process personal information from clients, employees, and other parties, compliance with these regulations is an important part of their daily operations (National Privacy Commission, 2022).

Many law firms use different software applications to manage cybersecurity and legal compliance. For example, one system may be used for antivirus protection, another for firewall monitoring, another for document storage, and another for compliance tracking. While these tools can perform their individual functions effectively, using multiple systems often creates challenges. Employees may need to switch between several applications, making work more time-consuming and increasing the possibility of missing security alerts, software updates, or compliance deadlines. Managing different platforms may also require additional training, higher

costs, and more effort from both legal and information technology personnel (International Organization for Standardization [ISO], 2022; Ries, 2023).

Because of these challenges, there is a growing need for an integrated platform that combines cybersecurity monitoring and legal compliance management into a single system. Having one centralized platform can help law firms monitor security threats, organize compliance activities, manage important documents, and generate reports more efficiently. It can also simplify daily operations, reduce human error, improve response to cybersecurity incidents, and support compliance with data privacy regulations (ISO, 2022).

Thus, this capstone project proposes the development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The proposed platform aims to provide law firms with a centralized solution that combines cybersecurity protection and legal compliance management in one system. It will help monitor security risks, organize compliance requirements, manage legal and security-related documents, and strengthen the protection of sensitive client information. By combining cybersecurity and legal compliance in one platform, the proposed system aims to help law firms work more efficiently and stay organized. It can help protect sensitive client information, make it easier to follow legal requirements, reduce cybersecurity risks, and improve the overall safety and reliability of the firm's daily operations. LEXSHIELD, law firms can better protect confidential information, respond to evolving cyber threats, and maintain the trust and confidence of their clients.

1.2 Statement of the Problem

The increasing use of digital technology in law firms has improved the way legal services are delivered and managed. Law firms now rely on computers, online storage, electronic

documents, and communication platforms to handle important information and communicate with clients. However, the use of these technologies also creates new challenges, especially in protecting confidential information and maintaining compliance with data privacy laws.

Many Philippine law firms still face difficulties in managing cybersecurity and legal compliance because existing solutions are often separated into different systems. Some firms use security tools to protect their devices and networks, while compliance tasks are handled manually through documents, spreadsheets, or separate applications. This makes it harder to monitor security risks, update compliance requirements, and respond quickly to possible threats.

Since law firms manage sensitive information such as client records, legal documents, financial details, and case-related information, the lack of proper security management may expose them to cyberattacks, unauthorized access, and data breaches. There is also a need for a centralized platform that can help law firms monitor security activities, manage compliance requirements, and make better decisions regarding data protection.

This study aims to address the following problems:

1. Existing law firms do not have a single platform that combines cybersecurity protection and legal compliance management in one system.
2. Current cybersecurity solutions mainly focus on protecting computer systems and networks but may not provide features that help law firms monitor legal and regulatory compliance requirements.
3. Confidential client information, including personal details, financial records, and legal documents, remains at risk due to possible cyber threats such as unauthorized access, malware attacks, and data breaches.

4. Law firms experience difficulties in tracking, updating, and maintaining compliance with the Data Privacy Act of 2012 and other related data protection requirements.
 5. Existing systems lack a centralized platform that provides real-time monitoring, security reports, risk assessment, and decision-support tools for effective cybersecurity and compliance management.
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1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system aims to provide law firms with a centralized platform that helps protect sensitive client information, improve cybersecurity practices, support compliance with legal requirements, and provide useful monitoring and management tools for administrators.

The proposed system is intended to help law firms manage security risks more effectively by combining cybersecurity features and compliance management processes into one platform. It will provide tools that can assist users in monitoring possible threats, organizing compliance activities, and maintaining the security of important information.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Identify and analyze the needs of law firms regarding cybersecurity and legal compliance management, including the problems and limitations of their current systems.

2. Design a secure, simple, and user-friendly interface that allows law firm users and administrators to easily access and manage the platform's features.
 3. Develop integrated modules that support cybersecurity monitoring, data privacy management, and compliance tracking based on applicable regulations.
 4. Implement monitoring features that can help detect possible security risks, track system activities, and provide updates regarding cybersecurity and compliance status.
 5. Develop an administrative dashboard that allows authorized users to manage system activities, generate reports, review security information, and support decision-making.
 6. Create a secure and centralized database for storing and managing important client and system information while maintaining data protection.
 7. Test and evaluate the developed platform based on security, usability, performance, reliability, and effectiveness in supporting cybersecurity and compliance management.
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1.4 Scope and Delimitation

This study focuses on the design and development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The system is created specifically to help law firms improve their management of cybersecurity and legal compliance through a centralized platform.

The proposed system will include important features such as user registration, user authentication, role-based access control, secure data storage, data encryption, compliance monitoring, security risk monitoring, threat detection, and an administrative dashboard. These features are intended to help law firms protect confidential information and manage cybersecurity-related activities more efficiently.

The compliance module of the system will focus on supporting requirements related to the Data Privacy Act of 2012 and other relevant data protection practices. The platform will assist users in organizing compliance information, monitoring security activities, and generating reports that may help administrators review the status of their security and compliance efforts.

However, this study has several limitations. The system will only be developed as a prototype for academic purposes and will not be fully deployed across all law firms in the Philippines as an example. It will not include complete integration with existing law firm software, external government systems, or third-party applications. The study will also not cover advanced artificial intelligence-based prediction systems, automated legal decision-making, or large-scale cybersecurity operations.

The effectiveness of the system will be evaluated based on testing results related to usability, functionality, security features, and overall performance within the scope of the developed prototype.

1.5 Significance of the Study

The proposed **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms** may provide benefits to the following groups:

- **Law Firms**
 - This study may help law firms improve the protection of confidential client information by providing a centralized platform for cybersecurity monitoring and compliance management. It may reduce the difficulty of managing different

security tools and help firms organize their data protection processes more effectively.

- **Administrators and IT Personnel**

- The system may assist administrators and IT personnel by providing tools for monitoring security activities, managing user access, reviewing reports, and identifying possible risks. It may also help them maintain better control over cybersecurity and compliance tasks.

- **Clients**

- Clients may benefit from improved protection of their personal and legal information. By using a platform focused on cybersecurity and compliance, law firms can strengthen their ability to protect client data and maintain client trust.

- **BSIT Students**

- This study may serve as a learning resource for BSIT students who are interested in software development, cybersecurity, database management, and system integration. It may provide an example of applying information technology concepts to solve real-world problems.

- **Future Researchers**

- Future researchers may use this study as a reference for developing similar systems related to cybersecurity, legal technology, data privacy, and compliance management. It may also provide ideas for improving or expanding integrated security platforms in different industries.
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1.6 Definition of Terms

- **Cybersecurity**
 - The process of protecting computers, networks, applications, and digital information from cyber threats, unauthorized access, and possible damage.
- **Legal Compliance**
 - The process of following required laws, regulations, policies, and standards that apply to an organization's activities and operations.
- **Data Privacy Act of 2012**
 - A Philippine law that provides rules and guidelines for the proper collection, processing, storage, and protection of personal information (Republic Act No. 10173, 2012).
- **Integrated Platform**
 - A software system that combines different features and functions into one centralized application instead of using separate systems.
- **Administrative Dashboard**
 - A system interface that allows administrators to monitor activities, manage users, review information, and generate reports.
- **Real-Time Monitoring**
 - The continuous process of observing system activities, security events, and possible threats as they happen.
- **Threat Detection**
 - The process of identifying possible cybersecurity risks, attacks, or suspicious activities within a system.

- **Data Encryption**

- A security method that converts information into a protected format to prevent unauthorized users from accessing it.

- **Role-Based Access Control**

- A security method that limits user access based on their assigned roles and responsibilities within the system.

- **Compliance Monitoring**

- The process of checking and tracking whether an organization follows required laws, policies, and security standards.
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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.0 Introduction

This chapter presents the foreign and local studies, related literature, and system concepts that are relevant to the development of **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. The main purpose of this chapter is to provide a strong foundation for the study by reviewing existing research, published articles, books, and other reliable sources related to cybersecurity, legal compliance, data privacy, and software development. These references help explain the current practices, challenges, and solutions that are connected to the proposed system.

Reviewing previous studies is important because it helps the researchers understand what has already been done and identify areas that still need improvement. It also provides ideas on the features, technologies, and methods that can be applied in developing the proposed platform. By examining different studies and literature, the researchers can compare various approaches and determine which ones are most suitable for the needs of law firms.

This chapter is divided into four main sections: foreign studies, local studies, related literature, and the synthesis of the reviewed works. These include topics such as cybersecurity, data privacy, information security, legal compliance, and software development. This section also discusses existing systems and tools that are related to the study and identifies their strengths and limitations. And, since law firms manage confidential client information and important legal documents, having an integrated system can help improve data security, support

compliance with legal requirements, reduce security risks, and make daily operations more efficient.

2.1 Foreign Studies

Foreign studies present ideas and findings from research done in other countries. These studies help explain how organizations handle cybersecurity, protect information, follow security rules, maintain privacy, and manage risks. They also provide useful examples, tools, and practices that can serve as references in developing the proposed cybersecurity compliance management system for law firms. For easier presentation and discussion, the foreign studies are organized into the following categories:

A. Cybersecurity Challenges and Protection Needs in Law Firms

“Law Firm Cybersecurity: The State of Preventative and Remedial Regulation Governing Data Breaches in the Legal Profession”

Tarr (2017)

Tarr (2017) explored the growing cybersecurity challenges faced by law firms and explained why legal professionals play an important role in protecting confidential client information. The paper described how law firms handle highly sensitive records, such as legal documents, privileged communications, financial information, and personal data, making them attractive targets for cybercriminals. It discussed how data breaches can affect client confidentiality, damage a firm's reputation, and lead to legal and financial consequences. The author emphasized that these risks can be reduced by adopting preventive cybersecurity measures, complying with applicable regulations, and preparing effective incident response plans. The study concluded that law firms should establish clear cybersecurity policies and strengthen their security practices to safeguard client information and maintain public trust in legal services.

“Cybersecurity in Accounting Research: Current State and Future Directions”

Haapamäki & Sihvonen (2019)

They examine the current condition of cybersecurity research in the accounting field. Their study reviewed 39 theoretical and empirical publications to identify the main topics that researchers have explored over the years. These topics included cybersecurity information sharing, cybersecurity investments, internal auditing and security controls, cybersecurity disclosures, and security threats and data breaches. The authors explained that cybersecurity is not limited to using advanced technologies or security software. Instead, organizations also need effective internal controls, proper risk management strategies, clear security policies, and active participation from management and employees to protect sensitive information. The review also showed that cyber threats continue to become more complex, creating new challenges for organizations that handle confidential financial and client data. Because of these changing risks, the authors emphasized the need for more research to develop better cybersecurity practices, improve organizational preparedness, and support stronger decision-making in cybersecurity management. Overall, the study provides a clear overview of existing cybersecurity research and offers a useful framework that researchers and organizations can use to strengthen cybersecurity programs, improve security governance, and better protect valuable information from cyber threats.

“Does Cybersecurity Regulation Reduce Corporate Data-Breach Risk?”

Wang et al. (2025)

They investigated the effectiveness of cybersecurity regulations in reducing corporate data-breach risks by analyzing data from 3,761 publicly listed firms in China. The researchers examined how regulatory requirements influence organizations' cybersecurity practices and whether compliance with these regulations contributes to stronger protection against cyber

incidents. The study focused on understanding the relationship between government-imposed cybersecurity policies and organizational efforts to safeguard sensitive information.

The findings showed that cybersecurity regulations significantly reduce the likelihood of corporate data breaches by encouraging organizations to strengthen their cybersecurity culture, enhance security governance, and increase investments in cybersecurity technologies and protective measures. The researchers also found that organizations subject to stronger regulatory oversight were more likely to implement effective security controls and improve their overall preparedness for cyber threats. The study concluded that cybersecurity regulations are most effective when organizations integrate compliance requirements into their overall cybersecurity strategy rather than treating them merely as legal obligations. These results demonstrate that legal compliance, sound governance, and proactive cybersecurity management work together to improve organizational resilience and protect confidential information from evolving cyber risks.

"Client Confidentiality as Data Security"

Perlin (2024)

With the increasing reliance of law firms on digital technologies, Perlin (2024) examined how the concept of attorney-client confidentiality has expanded beyond traditional methods of protecting information. The study explained that lawyers are now responsible not only for preventing unauthorized disclosure of client information but also for ensuring that digital platforms, cloud services, and electronic communication systems are secured against possible cyber threats.

The research emphasized that law firms handle highly sensitive information, including legal documents, case records, confidential discussions, and personal data, which makes cybersecurity an important part of protecting client relationships and trust. Perlin discussed how

the use of modern technologies introduces new security challenges, such as unauthorized access, data exposure, and vulnerabilities in digital storage systems. The author highlighted the need for law firms to implement effective information security practices, make responsible technology choices, and establish procedures that minimize the impact of cybersecurity incidents. The study concluded that maintaining client confidentiality in today's digital environment requires lawyers to view cybersecurity as a core professional obligation and not merely as a technical issue managed by information technology personnel.

"When Should Law Firms Notify Clients About Data Breaches?"

Goldsmith, Standish, & Temkin (2021)

Goldsmith, Standish, and Temkin (2021) examined the challenges faced by law firms in responding to cybersecurity incidents involving the exposure of confidential client information. The study discussed how legal organizations have become attractive targets for cybercriminals because they handle valuable data, including client records, case documents, litigation files, and private communications. The authors highlighted that data breaches, ransomware attacks, and unauthorized access incidents can create serious risks for both law firms and their clients.

The research focused on the responsibility of law firms to determine when clients should be informed after a cybersecurity incident occurs. The authors explained that firms must assess factors such as the severity of the breach, the type of information compromised, and the potential effects on affected clients before deciding on proper notification actions. They emphasized the importance of having established cybersecurity policies, incident response plans, and communication procedures to manage security incidents effectively.

The study concluded that cybersecurity readiness is an essential part of modern legal practice. Proper handling of data breaches helps law firms protect client confidentiality, maintain

professional obligations, preserve client trust, and reduce the negative effects of cyberattacks on their operations and reputation.

B. Information Security Governance, Privacy, and Cybersecurity Frameworks

“Privacy by Design: The 7 Foundational Principles”

Cavoukian (2011)

The author emphasized that protecting personal information should begin long before an information system is completed. Through the concept of Privacy by Design, Cavoukian (2011) explained that privacy should be included in every stage of system planning, design, and development instead of being treated as an additional feature after problems arise. The paper described seven principles that guide organizations in creating systems that protect users' personal information. These principles include making privacy the default setting, ensuring transparency in data handling, implementing effective security safeguards, and giving users greater control over their information. By following these principles, organizations can reduce privacy risks, improve the security of their systems, and strengthen the confidence of users in how their personal data is managed.

“Privacy by Design: The 7 Foundational Principles”

von Solms & von Solms (2009)

The paper explored the importance of information security governance in organizations and explained why it should be considered an essential part of overall business management. The authors pointed out that cybersecurity is not only the responsibility of the information technology department but also a strategic concern that requires the involvement of top management and other key decision-makers. The paper discussed how leadership, clear security policies, effective risk management, and accountability all contribute to protecting an organization's information assets. It also emphasized that organizations are more likely to

achieve strong cybersecurity when their security objectives are aligned with their overall business goals and compliance requirements. By treating information security as part of organizational governance, companies can improve decision-making, strengthen their security practices, and better protect sensitive information from existing and emerging cyber threats.

“The Need for Information Security Governance: An Enterprise Perspective”

Posthumus & von Solms (2004)

According to Posthumus and von Solms (2004), organizations need more than technical security tools to effectively protect their information and manage today's cybersecurity challenges. The paper explained that technologies such as firewalls, antivirus software, and encryption are important, but they cannot provide complete protection when used on their own. The authors argued that organizations should also establish formal information security governance that includes clear policies, well-defined procedures, effective risk management, and accountability at every level. These governance practices help ensure that security measures are consistently applied and supported by both management and employees. The paper also pointed out that having a structured governance framework allows organizations to protect valuable information assets while meeting business objectives and complying with legal and regulatory requirements. Overall, the authors emphasized that combining technical security with strong governance creates a more effective and organized approach to managing cybersecurity risks and protecting sensitive information.

“Information Security Governance Challenges and Critical Success Factors:

Systematic Review”

Alshaikh (2020)

This systematic review examined the different challenges organizations face when implementing information security governance. The researchers analyzed previous studies to identify the most common issues that affect the successful management of information security.

The review found that many organizations struggle with implementing security policies consistently, managing cybersecurity risks effectively, aligning security practices with legal and regulatory compliance requirements, and maintaining continuous monitoring of their information systems. It also highlighted that these challenges are often influenced by organizational factors such as limited resources, lack of management support, insufficient employee awareness, and rapidly changing cyber threats. Based on the findings, the study concluded that effective cybersecurity governance cannot rely on technology alone. Instead, it requires well-defined governance frameworks, strong commitment from organizational leaders, clear policies and responsibilities, regular risk assessments, continuous monitoring, and ongoing evaluation to ensure that security measures remain effective and responsive to emerging threats. Overall, the review emphasizes that a structured and proactive approach to information security governance is essential for protecting organizational information assets and improving an organization's overall cybersecurity posture.

"Cybersecurity Frameworks and Their Role in Improving Organizational Security"

Calder & Watkins (2019)

The authors explained that cybersecurity frameworks provide organizations with a structured method for improving their security management practices. Calder and Watkins (2019) discussed how standards such as ISO/IEC 27001 help organizations identify possible security risks, establish appropriate controls, monitor the effectiveness of security measures, and prepare for regulatory compliance. The paper emphasized that organizations with well-defined cybersecurity frameworks are better equipped to manage security challenges and respond to changing threats. It also noted that these frameworks encourage continuous assessment and improvement, allowing organizations to strengthen their overall security posture over time. The authors concluded that adopting standardized cybersecurity frameworks supports consistent

security practices, improves organizational readiness, and enhances the protection of valuable information assets.

"Assessing Frameworks for Eliciting Privacy & Security Requirements from Laws and Regulations"

Oluwafemi Olukoya (2022)

The study evaluates different frameworks used to identify privacy and security requirements from laws and regulations for software development. As organizations collect and manage more personal data, complying with privacy laws has become increasingly important. However, converting legal requirements into software specifications remains challenging because legal documents are often complex and difficult to interpret.

The research reviews existing requirement elicitation frameworks and examines whether they can be applied beyond commonly studied regulations such as the GDPR and HIPAA. To assess their effectiveness, the study uses the Nigerian Data Protection Regulation (NDPR) as a case study and compares the extracted requirements with the privacy practices of FinTech organizations.

The findings show that current frameworks are useful for identifying legal privacy and security requirements but have limitations when applied to different legal systems. The study recommends developing more flexible and standardized frameworks that can help software engineers create systems that are both secure and compliant with various privacy laws.

"Arguing Regulatory Compliance of Software Requirements"

G. Siena, A. Perini, A. Susi, & J. Mylopoulos (2013)

The study introduces a framework that helps software developers determine whether software requirements comply with legal and regulatory obligations during the software

development process. It uses argumentation techniques to establish clear relationships between legal regulations and software requirements, allowing developers to justify why specific system features satisfy legal provisions. This structured approach makes compliance analysis more transparent, organized, and easier to verify.

Instead of treating regulatory compliance as a task performed after software has been developed, the framework integrates legal requirements into the early stages of system design. By identifying possible compliance issues during requirements engineering, developers can resolve legal conflicts before implementation, reducing the need for costly changes later in the development process. The framework also supports documenting compliance decisions, making it easier for organizations to demonstrate that their software follows applicable laws and regulations.

Overall, the study shows that incorporating legal compliance into requirements engineering improves software quality, minimizes legal and regulatory risks, and helps organizations develop systems that are both functional and compliant with relevant legal standards.

"An Approach Based on Citation Analysis to Support Effective Handling of Regulatory Compliance"

M. Ghanavati, D. Amyot, & L. Peyton (2011)

This research study presents a method that helps organizations manage and understand legal and regulatory requirements more effectively by using citation analysis. Since laws and regulations often refer to other legal documents, organizations may find it difficult to identify how different rules are connected and how changes in one regulation can affect others. The

proposed approach analyzes these legal references to create clear links between related regulations, making them easier to organize, understand, and interpret.

The method also helps track updates and changes in legal documents, allowing organizations to identify which compliance requirements may be affected and what actions may be needed to remain compliant. It provides a structured way to examine legal relationships, making it easier to monitor regulatory changes and assess their impact on organizational policies and software requirements. This approach can also improve communication between legal experts and software developers by presenting regulatory information in a more organized and understandable way.

Thus, the study shows that using citation analysis can improve the management of regulatory compliance, reduce the risk of missing important legal obligations, and help organizations maintain compliance as laws and regulations continue to evolve.

"Governing the Fiduciary Relationship in Information Security Services"

Kankanhalli, Teo, Tan, & Wei (2016)

According to Kankanhalli, Teo, Tan, and Wei (2016) examined how governance mechanisms influence the relationship between organizations and information security service providers. The study focused on the fiduciary nature of information security services, emphasizing that organizations must establish appropriate governance structures to ensure that service providers act in the best interests of their clients. The authors explained that effective governance promotes accountability, clearly defines roles and responsibilities, and builds trust between organizations and external security providers.

The researchers further discussed that successful information security governance extends beyond technical safeguards by incorporating managerial oversight, contractual

controls, and continuous collaboration between stakeholders. Their findings showed that organizations with well-defined governance mechanisms are better able to align cybersecurity initiatives with business objectives, improve decision-making, and enhance the protection of organizational information assets. The study concluded that strong governance frameworks contribute to higher information security performance by fostering accountability, supporting compliance efforts, and ensuring that security responsibilities are effectively managed across the organization.

C. Cybersecurity Compliance Platforms and Integrated Security Systems

" A Customizable Web Platform to Manage Standards Compliance of Information Security and Cybersecurity Auditing "

Antunes, Maximiano, & Gomes (2022)

This study developed a customizable web-based platform designed to improve the management of cybersecurity auditing and compliance requirements within organizations. The researchers explained that traditional compliance processes are often fragmented, time-consuming, and heavily dependent on manual documentation, making it difficult for organizations to monitor compliance activities and maintain accurate records. These challenges can lead to delays in audits, inconsistent reporting, and difficulty in identifying security issues. To address these problems, the proposed platform integrates several important functions into a single system, including security auditing, compliance evaluation, documentation management, and corrective action tracking. By centralizing these processes, the platform enables organizations to organize compliance information more efficiently, monitor audit progress, and respond more quickly to identified security gaps.

The results show that using a centralized web-based platform improves the efficiency of cybersecurity audits, reduces the administrative burden associated with manual processes, and

helps organizations detect security weaknesses more effectively. Overall, the study demonstrates that integrated compliance management systems can strengthen cybersecurity governance, support regulatory compliance, and improve an organization's ability to maintain a secure and well-managed information security environment.

" Cyber Security and Compliance Management Through a Single Integrated Platform "

Jagtap, Pagar, & Meshram (2018)

According to Jagtap, Pagar, and Meshram (2018), organizations can improve their cybersecurity by using a single platform that combines different security functions into one system. The paper presented an integrated cybersecurity platform that includes security assessment, vulnerability analysis, auditing, and compliance management. The authors explained that many organizations rely on separate security tools to perform these tasks, which often leads to fragmented monitoring, duplicated work, and delays in responding to security issues. They pointed out that managing multiple systems can make cybersecurity operations more difficult and less efficient. By bringing these functions together in one centralized platform, organizations can monitor their security environment more effectively, simplify compliance activities, and improve their ability to detect and respond to potential threats. Overall, the paper emphasized that an integrated platform helps strengthen cybersecurity management while making security operations more organized and efficient.

" Compliance Management Systems: Do They Make a Difference? "

Coglianesse & Nash (2021)

The research emphasized that compliance management systems are valuable tools for helping organizations meet legal and regulatory requirements in a more organized and efficient manner. Coglianese and Nash (2021) explained that these systems assist organizations in identifying compliance obligations, assigning responsibilities, monitoring performance, and

correcting problems before they become more serious. The paper pointed out that technology improves the efficiency of compliance activities by making information easier to manage and helping organizations monitor their progress more effectively. At the same time, the authors stressed that successful compliance depends on strong governance, clear policies, and active support from organizational leaders. Overall, the paper showed that combining technology with proper governance strengthens accountability, improves risk management, and helps organizations maintain consistent compliance with applicable laws and standards.

" Holistic Information Security Management and Compliance Framework "

Grigaliūnas et al. (2024)

The authors emphasized the importance of adopting a holistic approach to information security management by combining cybersecurity controls, governance practices, and compliance requirements into a single framework. Grigaliūnas et al. (2024) explained that many organizations manage these areas separately, which can create gaps in security and increase the risk of cyber threats and regulatory issues. The paper pointed out that an integrated framework helps organizations coordinate their security activities more effectively by ensuring that technical controls, organizational policies, and compliance efforts work together.

The authors also noted that this approach improves communication among different departments, supports better decision-making, and strengthens the organization's ability to manage security risks. Overall, the paper concluded that a unified information security framework helps organizations improve regulatory compliance, protect sensitive information, and establish a more consistent and effective cybersecurity management system.

"Applying Information Security Governance"

Moulton & Coles (2003)

The study by Moulton and Coles (2003) explored the role of information security governance as an essential component of effective organizational management. As businesses

became increasingly dependent on information systems and faced stricter corporate governance requirements, the authors argued that information security should no longer be viewed solely as a technical concern but as a strategic management responsibility. They explained that organizations need a structured governance framework that clearly defines security policies, roles, responsibilities, and accountability to effectively manage information security risks.

In addition, the study emphasized that successful information security governance requires the integration of risk management, internal controls, and continuous monitoring into an organization's overall governance structure. The authors noted that senior management plays a critical role in establishing security objectives, allocating resources, and ensuring compliance with organizational policies and regulatory requirements. By embedding information security into business decision-making processes, organizations can strengthen the protection of their information assets, improve operational efficiency, reduce security vulnerabilities, and enhance stakeholder confidence. The study concluded that a governance-based approach provides organizations with a more sustainable and effective way to manage information security than relying solely on technical security solutions.

D. Data Privacy, Regulatory Compliance, and Legal Information Protection

" The EU General Data Protection Regulation (GDPR): A Practical Guide "

Paul Voigt and Axel von dem Bussche (2017)

This research examined the requirements and implications of the General Data Protection Regulation (GDPR) for organizations handling personal information. The authors explained that organizations must implement appropriate technical and organizational security measures to protect personal data and demonstrate compliance with privacy regulations. The study emphasized accountability, documentation, data protection policies, and continuous compliance monitoring as essential practices for organizations managing sensitive information.

" Cybersecurity Threats and Vulnerabilities: A Systematic Review "

Al-Rimy, Maarof, & Shaid (2019)

This systematic review examined the different cybersecurity threats, weaknesses, and attack techniques that affect organizations in the digital environment. The researchers found that organizations continue to experience growing security risks caused by threats such as malware, phishing attacks, unauthorized access, and data breaches. The study highlighted that relying only on reactive measures is not enough to protect sensitive information. Instead, organizations should implement proactive security practices, including continuous monitoring, regular vulnerability assessments, strong security controls, and effective risk management strategies. The research concluded that adopting a comprehensive cybersecurity approach is necessary to improve information protection and reduce the impact of evolving cyber threats.

" Blockchain's Roles in Strengthening Cybersecurity and Protecting Privacy "

Nir Kshetri (2017)

The research highlighted the growing importance of emerging technologies in improving cybersecurity and protecting sensitive information. Kshetri (2017) explained that organizations can strengthen their security by adopting secure information management approaches that help maintain data integrity, improve authentication, and reduce the risk of unauthorized access. The paper pointed out that traditional security methods alone may no longer be enough as cyber threats continue to become more advanced. For this reason, organizations are encouraged to implement stronger and more innovative security mechanisms that can better protect digital information. The author also emphasized that improving cybersecurity is essential for maintaining trust, safeguarding privacy, and ensuring the secure use of digital technologies. By adopting these approaches, organizations can reduce security risks and improve the overall protection of their information systems.

“Legal Policies Failing on Data Breaches? An Empirical Study of U.S. Information Security Law Implementations”

Zhang et al. (2023)

Zhang et al. (2023) investigated whether information security laws in the United States have been effective in reducing the occurrence of data breaches. Recognizing that governments have introduced and revised numerous legal policies to strengthen data protection, the researchers examined panel data on reported data breaches in the United States from 2005 to 2019 to assess whether these regulations have achieved their intended purpose. The study also evaluated the effectiveness of legal penalties in discouraging organizations from neglecting information security responsibilities.

Furthermore, the researchers sought to determine whether existing legal frameworks serve as effective deterrents against cybersecurity incidents or whether additional policy improvements are necessary. By analyzing the relationship between legal enforcement and the frequency of data breaches, the study provides empirical evidence on the strengths and limitations of current information security laws. The findings offer valuable insights for policymakers and organizations in developing more effective cybersecurity regulations, strengthening compliance mechanisms, and enhancing the protection of sensitive information against evolving cyber threats.

“Challenges and Enablers for GDPR Compliance: Systematic Literature Review and Future Research Directions”

Zaguir, Magalhaes, & Spinola (2024)

According to Zaguir, Magalhaes, and Spinola (2024), organizations face several challenges when trying to comply with the General Data Protection Regulation (GDPR). Through a systematic review of previous studies, the researchers identified common issues such as the

need to change organizational processes, limited technological capabilities, lack of knowledge about compliance requirements, and difficulties in turning legal obligations into practical security measures. These challenges make it difficult for many organizations to fully implement GDPR requirements.

The study also explains that GDPR compliance is not only about following legal rules but also about building strong privacy practices within the organization. This includes improving information governance, applying appropriate security controls, and continuously monitoring how personal data is managed. The researchers found that organizations are more successful when they use structured compliance frameworks, enterprise architecture management, and effective information governance to support their compliance efforts.

Thus, the findings show that combining legal compliance with cybersecurity strategies helps organizations strengthen data protection and reduce privacy risks. The researchers emphasize the value of integrated compliance solutions that assist organizations in managing privacy obligations, monitoring security practices, and maintaining accountability.

"Are We There Yet? Understanding the Challenges Faced in Complying with the General Data Protection Regulation (GDPR)"

Sirur, Nurse, & Webb (2018)

Sirur, Nurse, and Webb (2018) explained that many organizations experience practical difficulties while trying to comply with the General Data Protection Regulation (GDPR). Their study examined how organizations understand GDPR requirements, make the necessary operational changes, and determine whether their data protection practices meet the required standards. The research focused on the real challenges encountered during the implementation process.

The researchers found that organizations often struggle because GDPR contains complex legal requirements that are not always easy to interpret. Many also have difficulty identifying and managing personal data, understanding their legal responsibilities, and creating effective methods to verify compliance. In addition, organizations with limited privacy knowledge and weak data management practices face greater challenges in meeting GDPR requirements.

The study concluded that successful GDPR compliance requires organizations to adopt structured approaches for managing personal information, strengthening cybersecurity practices, and maintaining clear visibility over their data processing activities. Continuous monitoring, proper documentation, and good security awareness also play important roles in maintaining compliance. These findings provide useful guidance for developing systems that support automated compliance monitoring, privacy management, and the secure handling of confidential information.

"The General Data Protection Regulation: Requirements, Architectures, and Constraints"

Hjerppe, Ruohonen, & Leppänen (2019)

In their study, Hjerppe, Ruohonen, and Leppänen (2019) discussed how the General Data Protection Regulation (GDPR) affects software development and system architecture. The researchers explored ways organizations can translate legal privacy requirements into technical system requirements that can be implemented during software development. Their study highlights the importance of designing information systems that support privacy and security from the beginning.

The researchers identified several key GDPR requirements that software systems should include, such as protecting personal data, providing user privacy controls, ensuring transparency, maintaining accountability, and supporting secure data processing. They also

found that many organizations, especially small and medium-sized enterprises, face challenges because of limited financial resources, technical expertise, and system development capabilities needed to build privacy-compliant architectures. The findings emphasize that privacy and security should be considered during the early planning and design stages of software development instead of being added after the system has already been built. Integrating legal requirements into the system development process helps organizations create more secure, reliable, and compliant information systems.

"The General Data Protection Regulation and the Rise of Certification as a Regulatory Instrument"

Lachaud (2018)

Lachaud (2018) examined how certification mechanisms under the General Data Protection Regulation (GDPR) can help organizations demonstrate compliance with data protection requirements. The study analyzed how certification serves as a practical tool for showing that an organization's data processing activities follow privacy regulations and accepted security standards. It also explored how certification promotes accountability and encourages organizations to improve their privacy management practices.

The study explains that traditional regulatory approaches mainly depend on government enforcement after organizations violate the law. In contrast, certification encourages organizations to maintain compliance continuously by regularly reviewing and improving their privacy and security measures. The author also noted that certification helps organizations build trust with customers, business partners, and regulators by showing their commitment to protecting personal information responsibly.

The study revealed the importance of using compliance management tools that can document security practices, monitor privacy requirements, maintain proper records, and provide

evidence of compliance whenever necessary. The study suggests that integrating cybersecurity management with legal compliance functions enables organizations to manage their responsibilities more effectively. These findings support the development of integrated platforms that help organizations maintain both cybersecurity controls and legal compliance in an organized and efficient manner.

E. Risk Management, Threat Detection, and Incident Response

" How Integration of Cyber Security Management and Incident Response Enables Organizational Learning "

Ahmad, Maynard, & Shanks (2020)

The paper explored how integrating cybersecurity management with incident response processes can help organizations become more resilient against cyber threats. Ahmad, Maynard, and Shanks (2020) explained that organizations are better prepared to respond to security incidents when they have a structured process for detecting, documenting, analyzing, and managing cybersecurity events. The authors pointed out that every security incident provides valuable lessons that can be used to improve future security strategies and reduce the likelihood of similar attacks. They also emphasized that incident response should not only focus on recovering from an attack but should also support continuous learning and improvement. By reviewing past incidents and applying the lessons learned, organizations can strengthen their cybersecurity capabilities, improve decision-making, and respond more effectively to future threats. Overall, the paper shows that combining cybersecurity management with incident response creates a stronger and more proactive approach to protecting organizational information and systems.

" Information Security Management Needs More Holistic Approach: A Literature Review "

Soomro, Shah, & Ahmed (2016)

The paper examined the importance of adopting a holistic approach to information security management in organizations. Soomro, Shah, and Ahmed (2016) explained that effective cybersecurity cannot be achieved by relying on technology alone. Instead, organizations need to combine technical security measures with well-defined policies, employee awareness programs, and organizational processes to build a stronger security environment. The authors pointed out that each of these components plays an important role in protecting information assets and reducing the risk of cyber threats. They also emphasized that when technology, people, and organizational practices work together, organizations are better prepared to prevent security incidents and respond effectively when they occur. The paper concluded that an integrated approach to information security management helps organizations strengthen their overall cybersecurity, improve risk management, and ensure the continuous protection of sensitive information in an increasingly digital environment.

"Cybersecurity Risk Management: A Review of Current Challenges and Future Directions"

Amin, T., Ahmad, A., & Maynard, S. B. (2019)

According to Amin, Ahmad, and Maynard (2019), organizations face increasing challenges in managing cybersecurity risks because digital technologies and cyber threats continue to evolve. The study reviewed different cybersecurity risk management approaches and examined how organizations identify, assess, and respond to security threats. The researchers found that traditional risk management methods are often no longer enough since cyberattacks have become more advanced, frequent, and difficult to predict.

The study explains that organizations need structured risk management frameworks that include threat identification, vulnerability assessment, security controls, continuous monitoring,

and incident response planning. The authors also emphasized that effective cybersecurity risk management is not limited to using technical security tools. Organizations must also establish strong governance, develop clear security policies, and improve employee awareness to strengthen their overall security posture. Those findings suggest that organizations can improve their ability to prevent and respond to cyber threats by combining risk assessment with continuous threat monitoring and proactive response strategies. The researchers also highlighted the importance of systems that support ongoing security evaluation, risk tracking, and incident management.

"A Cybersecurity Risk Management Framework for Organizations"

Saeed, H., Yousafzai, S., & Gill, A. (2021)

Saeed, Yousafzai, and Gill (2021) proposed a cybersecurity risk management framework that helps organizations identify, assess, and manage cybersecurity threats more effectively. The researchers explained that organizations often struggle to understand their security weaknesses because cyber risks involve many factors, including technology, employee behavior, and organizational processes. These challenges make it difficult to develop effective security strategies without a structured approach.

The framework introduced in the study combines important cybersecurity activities such as risk identification, threat analysis, vulnerability assessment, mitigation planning, and continuous security improvement. The authors emphasized that organizations should move away from reacting only after security incidents occur and instead adopt proactive risk management strategies that reduce the chances of future cyberattacks.

"Cybersecurity Incident Response: A Systematic Literature Review"

Ahmad, A., Maynard, S. B., & Shanks, G. (2015)

In their study, Ahmad, Maynard, and Shanks (2015) examined how organizations manage cybersecurity incidents and improve their ability to respond to security threats. Through a systematic review of existing studies, the researchers analyzed different incident response practices and identified the key factors that contribute to effective cybersecurity incident management. Their review focused on how organizations can strengthen their response capabilities before, during, and after security incidents.

The researchers found that successful incident response depends on having well-defined procedures, effective communication, trained personnel, and continuous improvement based on lessons learned from previous incidents. They also explained that incident response should not only focus on recovering from cyberattacks but should also include investigating the causes of incidents and implementing measures to prevent similar attacks in the future.

"Information Security Risk Management: A Review of the Literature"

Soomro, Z. A., Shah, M. H., & Ahmed, J. (2016)

Based on the review conducted by Soomro, Shah, and Ahmed (2016), information security risk management plays an important role in protecting organizational information from cybersecurity threats. The researchers examined different approaches used by organizations to manage information security risks and found that effective protection requires more than just technical security solutions. It also depends on proper planning, organizational policies, and continuous risk assessment.

The study explains that organizations should regularly identify security vulnerabilities, evaluate possible threats, and apply appropriate security controls to reduce risks. The authors also highlighted the importance of security awareness among employees, strong governance, and compliance with security standards to ensure that sensitive information remains protected from both internal and external threats.

"A Review of Cybersecurity Risk Management Frameworks"

Alsmadi, I., Al-Kabi, M., Alsmadi, H., & Al-Kharabsheh, A. (2018)

Alsmadi, Al-Kabi, Alsmadi, and Al-Kharabsheh (2018) reviewed different cybersecurity risk management frameworks that organizations use to identify, evaluate, and reduce cybersecurity risks. The researchers compared several existing frameworks and discussed how structured risk management approaches help organizations strengthen their overall cybersecurity practices. The study also examined how these frameworks support organizations in dealing with changing cyber threats.

The researchers explained that cybersecurity frameworks provide systematic methods for identifying vulnerabilities, assessing risks, implementing security controls, monitoring threats, and continuously improving security practices. They emphasized that because cyber threats constantly evolve, organizations should adopt flexible and adaptive risk management strategies that can respond to new and emerging security challenges. The findings show that effective cybersecurity risk management depends on continuous risk assessment, proactive threat monitoring, and close integration between technical security measures and organizational processes.

F. Cybersecurity Information Sharing, Collaboration, and Knowledge Management

" Cybersecurity Information Sharing Governance Structures: An Ecosystem of Diversity, Trust, and Tradeoffs "

Elaine M. Sedenberg & James X. Dempsey (2018)

According to Sedenberg and Dempsey (2018), cybersecurity information sharing plays an important role in helping organizations improve their ability to detect and respond to cyber threats. The paper explained that successful collaboration depends on trust between organizations, the protection of sensitive information, well-defined policies, and effective

information-sharing structures. The authors pointed out that organizations are more willing to share valuable security information when clear guidelines and governance practices are in place. They also emphasized that sharing threat information allows organizations to recognize potential risks more quickly, improve their understanding of emerging cyber threats, and strengthen their incident response capabilities. Overall, the paper suggests that building trust and establishing proper governance are essential for creating effective cybersecurity partnerships and improving organizational resilience against cyberattacks.

" Knowledge Management for Cybersecurity in Organizations "

Chen, Wang, & Zhang (2021)

The paper explored the relationship between cybersecurity and knowledge management practices in organizations. Chen, Wang, and Zhang (2021) explained that managing and sharing security-related knowledge is an important part of building a strong cybersecurity program. The authors discussed how properly documenting security policies, incident reports, lessons learned, and best practices allows organizations to respond more effectively to cyber threats. They pointed out that when employees have access to accurate and up-to-date security information, they are better prepared to recognize risks, make informed decisions, and follow appropriate security procedures. The paper also emphasized that effective knowledge management supports continuous learning and helps organizations improve their cybersecurity strategies over time. Overall, the authors concluded that sharing and maintaining cybersecurity knowledge strengthens an organization's ability to respond to threats, improve decision-making, and protect valuable information assets.

"Cybersecurity Information Sharing: An Overview of Challenges and Solutions"

T. Sommestad, H. Karlzén, & J. Hallberg (2019)

According to Sommestad, Karlzén, and Hallberg (2019), sharing cybersecurity information among organizations plays an important role in improving protection against cyber

threats. Their study examined the common challenges and possible solutions related to exchanging information about cyberattacks, security vulnerabilities, and threat patterns. The researchers explained that organizations can strengthen their security by learning from one another, but many are still reluctant to share information because of concerns about confidentiality, privacy, legal responsibilities, and the possibility that sensitive data may be misused.

The study also reviewed different methods used to support cybersecurity information sharing, including technical platforms, information-sharing communities, and governance mechanisms. The authors emphasized that successful collaboration depends not only on technology but also on clear policies, trusted relationships, and strong organizational commitment to information sharing.

"A Survey of Cybersecurity Information Sharing: Challenges, Solutions, and Future Directions"

Y. Liu, X. Liu, & H. Zhang (2020)

Liu, Liu, and Zhang (2020) conducted a survey that explored the current practices, challenges, and future directions of cybersecurity information sharing. The researchers explained that organizations can improve their ability to defend against cyber threats by exchanging information about malicious activities, security incidents, and system vulnerabilities. However, they also found that many organizations face obstacles that limit effective information sharing, including lack of trust, privacy concerns, inconsistent data formats, and uncertainty about legal responsibilities.

The study reviewed different information-sharing models and technologies, such as cyber threat intelligence platforms, automated information exchange systems, and collaborative security networks. The authors explained that successful information sharing requires

standardized procedures, secure communication channels, and cooperation among organizations to ensure that information is exchanged safely and efficiently. The findings show that strengthening cybersecurity information-sharing mechanisms can improve threat detection, support faster incident response, and enhance overall cybersecurity management.

"Cyber Threat Intelligence Sharing: A Review of the Current State and Future Research Directions"

A. Tounsi & H. Rais (2018)

In their review, Tounsi and Rais (2018) discussed the importance of cyber threat intelligence sharing in helping organizations strengthen their cybersecurity. The study explained that cyber threat intelligence enables organizations to collect, analyze, and exchange information about cyber threats so they can prepare for attacks and respond more effectively. By sharing threat intelligence, organizations can identify attack patterns, recognize new risks, and make better security decisions.

The researchers examined several cyber threat intelligence frameworks and identified common challenges that affect information sharing. These include poor data quality, lack of trust between organizations, privacy concerns, and difficulties integrating information from multiple sources. The authors emphasized that effective intelligence sharing requires cooperation among organizations, government agencies, cybersecurity providers, and other stakeholders.

The conclusions suggest that improving cyber threat intelligence exchange helps organizations develop proactive cybersecurity strategies, detect threats more quickly, and strengthen their overall defense capabilities.

"Sharing Cybersecurity Information: A Review of the Challenges and Opportunities"

J. R. Cárdenas, R. González, & M. Fernández (2021)

Cárdenas, González, and Fernández (2021) explained that cybersecurity information sharing has become an important strategy for improving organizational security as cyber threats continue to grow in complexity. The researchers pointed out that it is becoming increasingly difficult for organizations to defend themselves against cyberattacks without collaborating with others. Sharing cybersecurity knowledge, threat intelligence, and security experiences helps organizations improve awareness, identify vulnerabilities, and respond more effectively to security incidents.

The study examined the technical, organizational, and legal factors that influence successful information sharing. The authors found that organizations need to establish trust, protect confidential information, define clear responsibilities, and implement secure information-sharing platforms to support effective collaboration. They also emphasized that good governance practices are necessary to ensure reliable and secure information exchange. The observation shows that cybersecurity information sharing is an essential part of modern cybersecurity strategies because it promotes collaboration, improves threat intelligence capabilities, and strengthens organizational resilience.

"Cybersecurity Knowledge Management: A Framework for Improving Organizational Security Awareness"

M. P. Gupta, R. Agrawal, & A. Sharma (2019)

Based on the study by Gupta, Agrawal, and Sharma (2019), effective knowledge management is an important factor in improving cybersecurity awareness and strengthening organizational security. The researchers explained that cybersecurity depends not only on

technical security controls but also on the knowledge, awareness, and ability of employees to recognize and respond to security risks. Organizations with well-informed employees are better prepared to prevent cybersecurity incidents caused by human error.

The study proposed that organizations should establish structured knowledge management practices, including documenting cybersecurity policies, sharing lessons learned from previous incidents, maintaining security records, and providing regular employee training. The authors emphasized that continuous knowledge sharing helps build a strong security culture and encourages employees to follow good cybersecurity practices.

The findings indicate that organizations with effective cybersecurity knowledge management are more capable of preventing security incidents, responding to cyber threats, and continuously improving their cybersecurity performance. The researchers concluded that knowledge management should be considered an essential part of cybersecurity governance and organizational resilience.

2.2 Local Studies

Local studies are research conducted in the Philippines by Filipino researchers or institutions that provide information about issues, practices, policies, and challenges within the country's context. For this study, local research was reviewed to better understand cybersecurity, data privacy, legal compliance, cybercrime laws, and information governance in the Philippines. These studies helped identify existing gaps and provided valuable information that supports the development of the proposed LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms. To make the review of related literature more organized and easier to understand, the local studies are grouped into the following categories:

A. Cybersecurity Studies

"Internet Security Awareness of Filipinos: A Survey Paper"

D. Omorog & R. P. Medina (2020)

As internet use continues to grow in the Philippines, understanding how people protect themselves online has become increasingly important. This research examined the level of internet security awareness among Filipinos by reviewing existing studies and survey results related to online behavior and cybersecurity practices. The researchers aimed to determine whether internet users have enough knowledge to recognize common cyber threats and whether they apply safe security practices while using online platforms.

The results showed that many Filipinos are already familiar with basic cybersecurity concepts and understand the importance of protecting their personal information. Most users know that passwords should be kept private and that suspicious websites and messages should be avoided. However, the research also revealed that many internet users still practice unsafe online habits. Some continue to use weak or repeated passwords, click links without checking their source, ignore software updates, and have difficulty identifying phishing emails and online scams. These behaviors increase the possibility of identity theft, unauthorized access to personal accounts, financial fraud, and other cyber-related crimes.

Another important point discussed in the research is that cybersecurity awareness should continue to improve because cyber threats become more advanced as technology develops. Simply knowing the basic security rules is not enough if users do not consistently apply them in their daily online activities. The researchers emphasized that schools, government agencies, businesses, and internet service providers should work together to promote cybersecurity awareness through seminars, training programs, public campaigns, and educational materials.

By increasing public knowledge and encouraging responsible online behavior, Filipinos can become better prepared to protect themselves from different forms of cyberattacks.

"The Current Landscape of Cybersecurity in the Philippines: A Bibliometric Analysis"

Wilber B. Sabado (2025)

Using bibliometric analysis, this research examined how cybersecurity studies in the Philippines have developed over the years. The author analyzed published journal articles, conference papers, and other academic works to determine the growth of cybersecurity research, identify the most common research topics, and examine publication trends. The purpose of the study was to provide a clear picture of the country's current cybersecurity research landscape and identify areas that still require further investigation.

The analysis showed that cybersecurity has become an increasingly active field of research in the Philippines because more organizations and individuals now rely on digital technologies for communication, business transactions, education, and public services. As cyber threats continue to increase, researchers have also become more interested in studying information security, cybercrime prevention, network security, digital forensics, and cybersecurity education. The growing number of publications indicates that cybersecurity is now recognized as an important field that supports national development and digital transformation.

Despite the increase in research output, the study also found that some important topics have received less attention. These include organizational cybersecurity, cyber defense planning, cloud security, artificial intelligence in cybersecurity, cybersecurity governance, incident response, and risk management. The author explained that these areas are becoming more important because organizations are adopting new technologies that also introduce new security risks.

The research concluded that continued investment in cybersecurity research is necessary to help organizations develop better security practices and strengthen national cybersecurity efforts. It also encouraged stronger collaboration among universities, government agencies, and private companies to produce more research that can address real-world cybersecurity problems and support evidence-based policymaking.

"A Comparative Study of the Philippines in a Global Cybersecurity Context and its Implications on Local Cybersecurity Practices"

David John Buenaventura Dacanay et al. (2024)

To understand the country's position in the global cybersecurity environment, the researchers compared the cybersecurity readiness of the Philippines with that of several other countries. Different factors were examined, including national cybersecurity strategies, government policies, workforce development, digital infrastructure, cybersecurity programs, and the ability to respond to cyber incidents. The comparison was intended to identify the country's strengths while recognizing areas that still need improvement.

The findings showed that the Philippines has made noticeable progress in improving its cybersecurity capabilities through the implementation of national cybersecurity plans, stronger government initiatives, and increased awareness of digital security. Various organizations have also invested in better security technologies and cybersecurity training to protect their digital systems from cyber threats.

Although significant improvements have been made, the study found that many challenges still remain. One major concern is the shortage of qualified cybersecurity professionals who can support organizations in managing cybersecurity risks. The researchers also pointed out that the implementation of cybersecurity policies is not always consistent across different sectors. In addition, many organizations continue to experience difficulties in keeping up

with rapidly changing cyber threats because attackers constantly develop more advanced methods of compromising systems and stealing sensitive information.

Based on these observations, the researchers emphasized that cybersecurity should be viewed as a continuous process rather than a one-time effort. Strong cooperation between the government, educational institutions, private companies, and cybersecurity professionals is necessary to improve national preparedness. The study concluded that regular policy improvements, workforce development, investment in cybersecurity technologies, and continuous public awareness programs are important steps toward building a safer digital environment in the Philippines.

"Internet Security and Privacy Awareness Among College Students in the Philippines"

Jim Alves-Foss & Baby Grace Llego (2024)

According to Alves-Foss and Llego (2024), cybersecurity awareness is an essential skill that helps college students protect their personal information and maintain their privacy while using digital devices and the internet. Their study explored whether the age at which students first started using technology, together with their level of technology exposure and frequency of internet use, had an influence on their understanding of cybersecurity and online privacy. To answer this, the researchers surveyed approximately 200 college students from several universities in the Philippines and evaluated both their cybersecurity awareness and their online security practices. After analyzing the collected data, they found that age, technology exposure, and internet usage did not have a significant relationship with the students' level of cybersecurity awareness. This means that students who had been using technology for many years were not necessarily more knowledgeable about cybersecurity than those with less experience.

The findings suggest that simply spending more time online does not automatically improve a person's understanding of online security. Because of this, the researchers recommended that colleges and universities provide cybersecurity education that is based on the specific learning needs of students. They emphasized that structured training and awareness programs are more effective than relying on technology experience alone to develop safe and responsible online behavior.

"Cybersecurity Skills in New Graduates: A Philippine Perspective"

John Paul P. Miranda, Marlon I. Tayag, & Joel D. Canlas (2025)

The research conducted by Miranda, Tayag, and Canlas (2025) focused on identifying the cybersecurity knowledge, technical competencies, and workplace skills that employers expect from newly graduated professionals in the Philippines. The researchers reviewed current industry demands and gathered information through surveys involving students, educators, and cybersecurity professionals to determine the skills that are most valuable in today's workforce. The findings revealed that employers highly value graduates who have a solid understanding of cybersecurity concepts and practical technical skills that can be applied to real security problems. However, the study also showed that technical knowledge alone is not enough to succeed in the field. Companies also give equal importance to soft skills such as communication, teamwork, critical thinking, problem-solving, adaptability, and ethical decision-making because these abilities help employees work effectively with others and respond to different cybersecurity challenges.

The researchers concluded that graduates who possess both technical and interpersonal skills are better prepared for professional responsibilities. Based on these findings, they recommended that Philippine higher education institutions strengthen their cybersecurity

curricula by combining technical training with activities that develop professional and personal skills to prepare students for future careers in cybersecurity.

"CyberSAFE: A Gamified Web Platform with Analytics to Promote Cybersecurity Literacy Among IT Students"

Alfredo L. Calimbo et al. (2025)

In their study, Calimbo and his fellow researchers (2025) developed CyberSAFE, a gamified web-based learning platform designed to improve the cybersecurity literacy of Information Technology students in the Philippines. The platform was created to make cybersecurity education more engaging by combining quizzes, interactive challenges, rewards, progress tracking, and learning analytics into the learning process. These features encouraged students to participate more actively in their lessons while giving instructors useful information about student progress and the topics where additional support was needed. During the implementation of the platform, students became more interested in cybersecurity lessons and showed a better understanding of online threats, password security, data privacy, phishing attacks, and other safe digital practices.

The researchers also observed noticeable improvements in both the students' cybersecurity awareness and their overall knowledge after completing the learning activities. The findings suggest that students learn more effectively when lessons are interactive and enjoyable. Based on the results, the researchers concluded that combining gamification with learning analytics can improve student engagement, strengthen cybersecurity knowledge, and encourage safer online behavior among Information Technology students.

"High Awareness, Persistent Vulnerability: Youth Cybersecurity Risk Perception and Behavior"

Arnel R. Rosano et al. (2026)

Rosano et al. (2026) examined the relationship between cybersecurity awareness, risk perception, and protective behavior among Filipino youth in contemporary digital environments. Using a cross-sectional mixed-methods research design, the study collected data from 403 college students to determine whether having knowledge of cybersecurity threats translates into safer online practices. The findings showed that most participants demonstrated a relatively high level of cybersecurity awareness, particularly regarding phishing attacks, online scams, identity theft, and data privacy. However, statistical analyses revealed only weak relationships between cybersecurity awareness and actual protective behaviors, such as verifying suspicious links and practicing safe public Wi-Fi usage. The researchers concluded that awareness alone does not consistently lead to secure online behavior because many young users continue to make risky decisions despite understanding common cyber threats. The study emphasized that cybersecurity vulnerability among youth is influenced not only by technical knowledge but also by behavioral habits, digital narratives, trust cues, and AI-generated misinformation.

Based on these findings, the researchers recommended that cybersecurity education should move beyond awareness campaigns by incorporating behavior-focused training, experiential learning activities, and realistic cybersecurity scenarios that encourage students to consistently apply safe online practices.

B. Data Privacy and Legal Compliance Studies

"Assessing Compliance of Philippine State Universities to the Data Privacy Act of 2012"

James V. Presbitero & Michelle Renee D. Ching (2018)

This research evaluated how Philippine state universities implement the requirements of the law by examining their institutional policies, privacy procedures, information security measures, and overall privacy management practices. The researchers wanted to determine

whether universities have established effective systems for protecting the personal information of students, faculty members, employees, and other stakeholders.

The evaluation revealed that many state universities have already taken important steps to comply with the Data Privacy Act. Several institutions have created privacy policies, assigned personnel responsible for data protection, implemented security measures, and developed procedures for collecting, storing, and processing personal information. These efforts demonstrate that many universities recognize the importance of safeguarding sensitive information and complying with national privacy regulations.

However, the study also identified several areas that require additional improvement. Some universities continue to experience difficulties in consistently implementing privacy policies, conducting regular employee training, maintaining proper documentation, performing privacy risk assessments, and strengthening information security management. According to the researchers, these challenges may affect an institution's ability to fully comply with the requirements of the law.

The study emphasized that data privacy compliance should be treated as a continuous organizational responsibility instead of a one-time requirement. Universities should regularly review their policies, improve employee awareness, strengthen security controls, and monitor their privacy practices to ensure that personal information remains protected. Continuous improvement will help educational institutions maintain compliance while building greater trust among students, employees, and the public.

"A Maturity Model Toolkit on Information Governance for Philippine Universities to Aid in Implementing Compliance to the Data Privacy Act of 2012"

Danny Cheng (2021)

This study examined the challenges that Philippine universities face in complying with the Data Privacy Act of 2012 (RA 10173). To address these challenges, the researcher developed an information governance maturity model toolkit specifically designed for higher education institutions. The toolkit provides a structured and step-by-step approach to improving data privacy compliance through practical resources, including guidance documents, self-assessment tools, Business Process Management Notation (BPMN) process flow examples, templates, and implementation advisories. These materials were created to help universities evaluate their current information governance practices and identify areas that require improvement. The toolkit was assessed using an iterative mixed-method approach with the participation of the National Privacy Commission, data protection officers, industry experts, and university stakeholders. The results showed that the toolkit is practical, easy to use, and effective in helping institutions translate regulatory requirements into everyday governance practices.

The researcher concluded that adopting a holistic and maturity-based information governance framework can strengthen data privacy compliance, improve information security, and provide a useful model that can also be adapted by organizations beyond the higher education sector.

"Understanding the Data Privacy Act of 2012: Safeguarding the Filipino's Digital Rights"

Pia Sarah Claveria, Marmelo V. Abante, & Florinda G. Vigonte (2025)

Recognizing that many universities need practical guidance in managing information responsibly, this research introduced an information governance maturity model toolkit to support compliance with the Data Privacy Act of 2012. The researcher designed the toolkit to help educational institutions evaluate the effectiveness of their current information governance practices and identify areas where improvements are needed. Rather than simply checking

whether an institution complies with legal requirements, the framework encourages continuous development of governance processes over time.

The maturity model assesses several important areas of information governance, including organizational leadership, policy development, records management, documentation, data handling procedures, privacy compliance, risk management, and information security controls. Through this structured assessment, universities can determine their current level of maturity, recognize weaknesses, and develop practical action plans to improve their governance practices. The toolkit also provides a clear basis for measuring progress as institutions strengthen their information management systems.

Another important contribution of the study is its emphasis on the broader value of information governance. Effective governance not only helps institutions comply with legal requirements but also improves accountability, transparency, operational efficiency, and decision-making. Proper governance ensures that information is collected, stored, shared, retained, and disposed of according to established policies while protecting sensitive data from unauthorized access or misuse.

The research concluded that implementing a structured information governance framework enables universities to continuously strengthen their privacy management practices and improve compliance with the Data Privacy Act of 2012. By regularly assessing and improving governance processes, institutions can better protect personal information, reduce organizational risks, and build a stronger culture of responsible information management.

"Perceptions of Students, Faculty and Administrative Staff on the Data Privacy Act: An Exploratory Study"

David Paul R. Ramos (2019)

To better understand how members of educational institutions view data privacy, this exploratory study examined the awareness and understanding of the Data Privacy Act among students, faculty members, and administrative staff. The research aimed to determine how familiar these groups are with their rights and responsibilities under the law and how their level of knowledge may affect the protection of personal information within the institution.

The findings revealed that participants had different levels of awareness regarding the Data Privacy Act. Some respondents were knowledgeable about the law and understood the importance of protecting personal information, while others had only limited knowledge of data privacy principles and legal responsibilities. The study showed that awareness often depended on an individual's role, experience, and exposure to privacy-related activities or training. Because of these differences, some members of the academic community may not fully understand the proper handling, collection, storage, and sharing of personal data.

The research emphasized that protecting personal information requires the participation of everyone in an educational institution, not only those assigned to privacy-related positions. For this reason, the researcher recommended conducting regular awareness campaigns, seminars, and training programs to improve knowledge of the Data Privacy Act. Strengthening privacy education can help students, faculty, and staff better understand their responsibilities, encourage compliance with privacy regulations, and promote a culture of responsible information management within educational institutions.

"RA 10173 and its Challenges to Philippine State Universities and Colleges"
Compliance Performance: The Case of Mindanao State University–General Santos City"
Lumer Jude P. Doce & Michelle Renee D. Ching (2018)

According to Doce and Ching (2018), higher education institutions have an important responsibility to protect the personal information of students, faculty, and staff by complying with

Republic Act No. 10173, also known as the Data Privacy Act of 2012. Their study examined how well Mindanao State University–General Santos City complied with the requirements of the law using a qualitative case study approach. The researchers assessed the university's existing data privacy practices and identified the challenges that affected the implementation of its privacy policies. The findings revealed that the university was only partially compliant with the requirements of the Data Privacy Act. The study showed that limited awareness of data privacy policies among employees, insufficient financial and technical resources, and the lack of institutional priority given to privacy programs were the major reasons for the university's low level of compliance. These challenges made it difficult for the institution to fully implement effective privacy measures and protect personal information.

Based on the findings, the researchers emphasized the need for stronger leadership support, continuous awareness campaigns, regular employee training, and better allocation of resources to help universities achieve full compliance with data privacy regulations and improve the protection of sensitive information.

"Philippine Data Privacy Law: Is it Implemented in a Private University Library, or Not?"

Rachelle Joy T. Lazo & David Cabonero (2021)

In their study, Lazo and Cabonero (2021) examined how the Philippine Data Privacy Act was being implemented in a private university library and evaluated the level of awareness among the people who regularly use and manage its services. The researchers applied a mixed-methods research design by collecting both quantitative and qualitative data to assess the library's existing privacy practices and determine how well library personnel and users understood data privacy principles. The findings showed that most stakeholders had a satisfactory level of awareness about data privacy and that the library generally complied with

the requirements of the Data Privacy Act. However, the researchers also identified several areas that required further improvement, particularly in strengthening privacy procedures, improving documentation, and increasing awareness through regular training programs and information campaigns.

The study concluded that protecting personal information in academic libraries requires continuous education, regular policy reviews, and consistent implementation of data privacy practices. These efforts can help ensure that libraries continue to provide secure and responsible management of personal data.

"Data Privacy Best Practices of a Local Higher Educational Institution: A Model for Governance"

Rushid Jay S. Sancon (2023)

Sancon (2023) explored the data privacy governance practices of Laguna State Polytechnic University to identify effective strategies that support compliance with the Data Privacy Act of 2012. The study focused on examining the university's privacy policies, organizational structure, and management practices to determine how these contributed to protecting personal information and strengthening data privacy governance. The findings revealed that the university had made significant improvements in its privacy program by establishing clear institutional policies, appointing a Data Protection Officer, conducting regular awareness and training activities, and implementing proper privacy management procedures. These initiatives helped the institution improve its ability to protect personal information while meeting the legal requirements of the Data Privacy Act.

The study also showed that having strong leadership support and well-defined governance practices played an important role in maintaining effective data privacy management. Based on the findings, the researcher proposed a governance model that can

serve as a useful guide for other higher education institutions in developing, improving, and sustaining their own data privacy programs and compliance efforts.

C. Cybercrime and Philippine Legal Framework Studies

"Safeguarding the Digital Nation: A Narrative Exploration of Roles, Objectives, Cybersecurity Challenges and COMELEAK Case"

Joana Marie Laroco, Florinda G. Vigonte, & Marmelo G. Abante (2025)

Using the COMELEAK incident as the main point of discussion, this paper explored the current state of cybersecurity governance in the Philippines and the challenges faced in protecting government information systems. The research discussed how one of the country's largest data breaches exposed the personal information of millions of registered voters and demonstrated the serious effects of weak cybersecurity practices. Through this case, the authors examined the responsibilities of government agencies in protecting sensitive information and responding to cyber incidents.

The paper also reviewed existing cybersecurity laws, national policies, and government programs that support cybersecurity and digital governance. While these legal measures provide a foundation for protecting information systems, the authors explained that effective implementation remains a major challenge. Factors such as limited technical resources, increasing cyber threats, coordination issues among government agencies, and rapidly changing technology continue to affect the country's ability to respond effectively to cyberattacks.

Another important point discussed in the study is the need for continuous improvement in national cybersecurity efforts. The authors emphasized that preventing future incidents requires stronger cooperation among government agencies, private organizations, cybersecurity professionals, and the public. They also highlighted the importance of regularly updating cybersecurity policies, strengthening security controls, improving employee awareness, and

investing in modern technologies. The study concluded that a proactive and well-coordinated cybersecurity strategy is necessary to better protect government information systems and maintain public trust in digital services.

"Safeguarding Cyberspace: A Comprehensive Analysis of the Cybercrime Prevention Act of 2012 (RA 10175) and Its Role in Philippine Digital Governance"

Antonio Suminig Jr., Marmelo V. Abante, & Florinda G. Vigonte (2025)

A comprehensive review of the Cybercrime Prevention Act of 2012 (Republic Act No. 10175) was conducted to examine its role in addressing cybercrime and supporting digital governance in the Philippines. The study explained the purpose of the law, the different cyber offenses it covers, and the responsibilities of government agencies responsible for enforcing its provisions. It also discussed how the law contributes to creating a safer online environment as more government services, businesses, and individuals depend on digital technologies.

The authors analyzed the implementation of the law by examining its strengths, limitations, and legal issues that have emerged since its enactment. The findings showed that the Cybercrime Prevention Act has become an important legal tool for investigating and prosecuting online crimes such as hacking, identity theft, online fraud, cyber libel, illegal access, and other cyber-related offenses. However, the study also recognized that enforcing cybercrime laws can be difficult because technology continues to evolve, allowing cybercriminals to develop new methods of attacking digital systems.

In addition, the paper discussed the importance of maintaining a balance between strengthening cybersecurity and protecting the rights of citizens. The authors emphasized that while effective law enforcement is necessary to reduce cybercrime, privacy rights, freedom of expression, and other fundamental rights must also be respected. They concluded that improving digital governance requires continuous updates to cybersecurity laws, stronger

cooperation among law enforcement agencies, better public awareness, and responsible use of digital technologies.

"Cybercrime Prevention Act of 2012 in Practice: Cybersecurity, Controversy, and the Future of Digital Rights in the Philippines"

Alyssa Brucal, Marmelo V. Abante, & Florinda G. Vigonte (2025)

According to Brucal, Abante, and Vigonte (2025), the Cybercrime Prevention Act of 2012 plays an important role in strengthening cybersecurity and addressing the growing number of cybercrimes in the Philippines. Their study reviewed how Republic Act No. 10175 has been implemented and examined its effects on cybersecurity, law enforcement, and the protection of digital rights. The researchers discussed how the law has helped government agencies investigate and prosecute different forms of cybercrime, including cyber libel, identity theft, online fraud, child exploitation, and other internet-related offenses. They also examined several well-known legal cases to demonstrate how the law has been applied in actual situations and how it has influenced the country's digital legal system. Despite these positive contributions, the study pointed out that some provisions of the law continue to raise concerns about freedom of expression, the right to privacy, and the balance between protecting public safety and preserving individual rights.

The researchers concluded that while RA 10175 remains an important legal framework for combating cybercrime, there is still a need for continuous policy improvements, fair and consistent implementation, and stronger safeguards for digital rights. They emphasized that these measures will help ensure that the law remains effective in addressing future cybersecurity challenges while protecting the rights of internet users in the Philippines.

D. Information Governance and Organizational Security

"National Government Agency's Compliance on Data Privacy Act of 2012: A Case Study"

Vicente A. Pitogo (2019)

Focusing on a Philippine national government agency, this case study evaluated how the organization complies with the requirements of the Data Privacy Act of 2012. The assessment examined the agency's privacy policies, security measures, information management practices, and overall compliance with the law. The purpose of the study was to determine whether the agency had established effective systems for protecting the personal information it collects, processes, and stores.

The evaluation showed that the agency had already implemented several privacy-related policies and security measures. However, the overall level of compliance was found to be only partial because of several existing challenges. Among the factors identified were limited employee awareness of data privacy responsibilities, insufficient resources to fully implement privacy programs, and difficulties in consistently applying privacy policies throughout the organization. These issues reduced the agency's ability to achieve full compliance with the requirements of the law.

The researcher explained that compliance with data privacy regulations requires continuous effort and strong organizational support. Employees should receive regular privacy training to improve their understanding of legal responsibilities and proper data handling practices. The study also recommended strengthening privacy policies, improving security controls, allocating sufficient financial and technical resources, and regularly monitoring compliance activities. According to the researcher, these improvements will help government agencies better protect personal information while maintaining public trust and meeting the standards required by the Data Privacy Act of 2012.

"Cross-border Data Regulation for Digital Platforms: Data Privacy and Security"

Atty. Aiken Larisa O. Serzo (2021)

The study discussed how organizations operating in different countries should comply with Philippine data privacy laws while managing personal information that may be stored, processed, or shared outside the country. The research highlighted the growing importance of international data protection as businesses continue to rely on cloud services, online applications, and global digital platforms.

The paper explained that organizations handling cross-border data must ensure that personal information remains protected throughout its entire lifecycle, regardless of where the data is processed or stored. It emphasized that compliance involves more than simply following legal requirements. Organizations should also establish clear data governance policies, implement strong security measures, manage privacy risks, and adopt responsible information management practices to prevent unauthorized access, data breaches, and misuse of personal information.

Another important point discussed in the study is the need for organizations to remain updated on both Philippine privacy laws and international data protection standards. Since different countries have different legal requirements, organizations must carefully manage cross-border data transfers to avoid legal and security issues. The paper concluded that effective information governance, continuous risk assessment, strong cybersecurity practices, and compliance with applicable privacy regulations are essential for protecting sensitive information and maintaining public confidence in digital services.

"Implementation of Digital Governance in the Philippine SUCs: Basis for an Enterprise-Level Information System Model"

Allen Paul Esteban, Keno Piad, Isagani Tano, Joseph Espino, Ace C. Lagman, Jayson Victoriano, & Jonilo Mababa (2024)

According to Esteban et al. (2024), digital governance has become an important part of improving the quality of services and decision-making in Philippine State Universities and Colleges (SUCs). Their study examined how digital governance can be strengthened through the development of an enterprise-level information system model that integrates different institutional systems into a single platform. The researchers explained that connecting various information systems can better support instruction, research, extension services, and administrative operations by allowing departments to work together more efficiently.

The findings showed that a unified information system improves communication among offices, supports faster and more accurate data sharing, reduces duplicate records, and enables administrators to make timely and informed decisions. The study also highlighted the importance of having standardized governance policies, system interoperability, and strong information security measures to protect institutional data from unauthorized access and cyber threats. Based on these findings, the researchers concluded that successful digital governance depends on committed leadership, organizational support, compliance with the Data Privacy Act of 2012, and the adoption of enterprise-wide governance frameworks that improve institutional performance and the delivery of public services.

"A Maturity Model Toolkit on Information Governance for Philippine Universities to Aid in Implementing Compliance to the Data Privacy Act of 2012 (RA 10173)"

Danny Cheng (2024)

The study conducted by Cheng (2024) introduced an information governance maturity model designed to help Philippine universities improve their compliance with the Data Privacy Act of 2012. The researcher developed a practical toolkit that allows higher education institutions

to assess their current governance practices, measure their level of maturity, and identify areas that require improvement. The toolkit covers several important aspects of information governance, including leadership, policy development, records management, privacy compliance, risk management, and information security. It was specifically designed to address the common governance challenges faced by universities while providing a clear and systematic approach for improving their information management practices. To ensure its effectiveness, the maturity model was validated through consultations with data protection officers, regulators, and university representatives, who confirmed its value in supporting institutional governance and privacy compliance.

Based on the findings, the researcher concluded that adopting a maturity-based governance framework can help universities strengthen data security, improve information management, enhance regulatory compliance, and build a more effective and sustainable governance system over time.

"Evaluating Public Information Systems: Enhancing Transparency, Efficiency, and Citizen Engagement in Digital Governance"

Rowena Gracilla, Florinda G. Vigonte, & Marmelo V. Abante (2025)

With the increasing use of digital technologies in government services, evaluating public information systems has become essential for improving digital governance. In this study, Gracilla, Vigonte, and Abante (2025) examined how Public Information Systems contribute to enhancing transparency, efficiency, accountability, and citizen participation in Philippine public institutions. The researchers identified five key areas that should be considered when evaluating these systems: accessibility and inclusiveness, security and data privacy, interoperability, system efficiency, and user experience. These factors were recognized as important in

developing reliable, secure, and user-friendly digital government services that meet the needs of both government agencies and the public.

The study also emphasized that effective governance policies should be supported by strong cybersecurity measures and data privacy practices to ensure that public information remains protected while improving the quality of government services. Based on their findings, the researchers concluded that regular system evaluation, continuous modernization, and ongoing improvements in information governance are necessary to strengthen public trust, improve institutional performance, and support the long-term digital transformation of public institutions in the Philippines.

E. Cybersecurity Awareness and Compliance Behavior

"Cybersecurity Awareness of College Students in a Private Higher Education Institution"

Buenaventura Alabab, Jonathan Cubol, Michael Angelo Pascual, Ednelyn Ubaldo, Divina Roma Sario, & Marjorie S. Velasco (2023)

According to Alabab et al. (2023), developing cybersecurity awareness among college students is important because they regularly use the internet for learning, communication, and other daily activities. Their study assessed the level of cybersecurity awareness among students in a private higher education institution in the Philippines using a descriptive-correlational research design. A total of 291 students participated in the survey, which measured their knowledge of phishing attacks, identity theft, password security, malware, and other common online threats. The findings showed that most students had a satisfactory level of cybersecurity awareness and understood the importance of protecting their personal information while using digital technologies. However, despite having this knowledge, many students remained

vulnerable to phishing attacks, identity theft, and other unsafe online practices. The researchers explained that awareness alone does not always lead to secure online behavior.

Based on these findings, they recommended strengthening institutional cybersecurity awareness programs, integrating cybersecurity topics into academic courses, and conducting regular training activities that encourage students to apply safe online practices in their everyday lives.

"Internet Security and Privacy Awareness Among College Students in the Philippines"

Jim Alves-Foss & Baby Grace Llego (2024)

In their study, Alves-Foss and Llego (2024) examined the relationship between students' exposure to technology and their level of cybersecurity awareness across several universities in the Philippines. Around 200 college students participated in the survey, which evaluated their understanding of password security, phishing attacks, online privacy, and other important cybersecurity concepts. The findings showed that most students had a good level of cybersecurity awareness and recognized the importance of protecting their personal information while using digital technologies. However, the researchers observed that students' level of awareness differed depending on their technological background and previous cybersecurity training. This suggests that simply having more experience with technology does not always result in stronger cybersecurity knowledge or safer online behavior.

The conclusion suggests the importance of continuously integrating cybersecurity education into higher education curricula through practical lessons, awareness campaigns, and hands-on activities to further improve students' digital safety and responsible online practices.

***"Cybersecurity Awareness, Data Privacy Practices, and Behavioral Compliance
Among College Students in a Provincial University in the Philippines"***

Enrique B. Picardal Jr. (2026)

The study conducted by Picardal (2026) explored the relationship between cybersecurity awareness, data privacy practices, and behavioral compliance among 200 college students in a provincial university in the Philippines. Using a quantitative descriptive-correlational research design, the researcher examined whether students who were knowledgeable about cybersecurity also practiced safe and responsible online behavior. The findings revealed that most students demonstrated a high level of cybersecurity awareness and had a good understanding of common online threats, cybersecurity risks, and the importance of protecting personal information. However, their actual data privacy practices and behavioral compliance were only at a moderate level, showing that many students did not consistently apply what they knew when using the internet in their daily activities.

The study also identified a significant but weak positive relationship between cybersecurity awareness and behavioral compliance, suggesting that awareness alone is not enough to guarantee secure online behavior. Based on these findings, the researcher recommended behavior-focused cybersecurity education, stronger institutional data privacy policies, and regular practical training to help students develop safer, more responsible, and consistent online habits.

***"Cybersecurity Awareness and Cybersecurity Behavior of High School Students in
Davao City: A Mediation Role of Perceived Behavioral Control"***

Kenneth Budiongan & Ramil Carballo (2024)

Using the Theory of Planned Behavior as its framework, Budiongan and Carballo (2024) examined the relationship between cybersecurity awareness and the online behavior of high school students in Davao City. The researchers also investigated whether perceived behavioral control, or the students' confidence in their ability to practice safe online behavior, influenced this relationship. The findings showed a significant positive relationship between cybersecurity awareness and secure online practices, indicating that students with higher awareness were more likely to behave responsibly while using digital technologies. In addition, students who believed they had greater control over their online actions were more confident in applying safe cybersecurity practices in their everyday internet use.

These results suggest that knowledge becomes more effective when students also have the confidence to apply what they have learned. Based on the findings, the researchers concluded that increasing awareness alone is not enough to promote safe online behavior. They recommended integrating cybersecurity education into secondary school curricula, strengthening digital citizenship programs, and providing practical learning activities that encourage students to practice responsible, safe, and ethical online behavior.

2.3 Related Literature

Related literature includes books, journals, research papers, articles, government publications, and other reliable sources that are related to the topic of the study. These sources provide useful information, ideas, and findings from previous researchers that help explain the research topic. It allows researchers to gather information from different sources and use it to support their research. Reading related literature helps researchers understand important concepts, compare the ideas of different authors, and see how previous studies are connected to their own research.

To make the discussion organized and easy to understand, the related literature in this study is divided into different topics. Each topic covers an important area that is connected to the research and presents literature from different sources that supports the study.

A. Cybersecurity

Cybersecurity has become an important part of every organization because many business activities now depend on digital systems and online services. As organizations continue to store confidential information electronically, they also become more vulnerable to cyber threats such as malware, phishing attacks, ransomware, unauthorized access, and data breaches. Because of these risks, organizations need to implement security measures that protect their systems and information from cybercriminals.

According to the National Institute of Standards and Technology (2024), cybersecurity is not limited to preventing attacks. It also involves identifying security risks, protecting information assets, detecting security incidents, responding to cyberattacks, and recovering from them. Following these practices helps organizations reduce security risks, maintain business operations, and protect confidential information. For law firms, cybersecurity is especially important because they manage legal documents, client records, contracts, and other sensitive information that must remain secure at all times.

According to the Cybersecurity and Infrastructure Security Agency (2023), cybersecurity is essential for protecting critical systems, sensitive information, and digital services against constantly evolving cyber threats. Organizations are encouraged to implement layered security measures such as multi-factor authentication, strong password policies, regular software updates, employee cybersecurity awareness training, and continuous monitoring. These security practices reduce the likelihood of successful cyberattacks and improve organizational resilience.

IBM (2024) reported that cyberattacks continue to increase in frequency and financial impact worldwide. Data breaches often result in operational disruption, financial losses, legal liabilities, and reputational damage. The report also found that organizations implementing strong security controls, artificial intelligence-assisted security monitoring, and incident response planning significantly reduce both the cost and recovery time associated with security incidents. These findings highlight the importance of investing in proactive cybersecurity measures.

Von Solms and Van Niekerk (2013) argued that cybersecurity extends beyond protecting computer systems and includes safeguarding people, organizational processes, information, and technology from cyber-related risks. They emphasized that effective cybersecurity requires cooperation between management, employees, and technical personnel. Organizations should establish comprehensive security strategies that address both technological vulnerabilities and human behavior to improve overall cyber resilience.

Craigen, Diakun-Thibault, and Purse (2014) defined cybersecurity as the collection of technologies, policies, processes, guidelines, and practices designed to protect cyberspace, users, organizations, and information systems from cyber threats. The authors emphasized that cybersecurity aims to preserve the confidentiality, integrity, and availability of information while ensuring that organizations can continue operating securely despite cyber risks. They further explained that cybersecurity is a multidisciplinary field involving technical, legal, organizational, and human factors that work together to protect digital environments. These concepts support the need for integrated cybersecurity solutions capable of protecting sensitive legal information handled by law firms.

B. Information Security

Information security focuses on protecting information from unauthorized access, disclosure, alteration, or destruction. It applies not only to electronic records but also to printed

documents and other forms of information used by an organization. Data privacy is concerned with protecting personal information and ensuring that it is collected, processed, stored, and shared responsibly. Organizations that collect personal information are expected to protect it against unauthorized access, disclosure, or misuse.

As mentioned by the National Privacy Commission (2016), organizations should implement organizational, physical, and technical security measures to safeguard personal data throughout its lifecycle. Good privacy practices help organizations comply with legal requirements while building trust among clients and stakeholders. Since law firms regularly process sensitive client information, maintaining strong data privacy practices is necessary to protect client rights and preserve confidentiality.

The National Institute of Standards and Technology (2020) explained that information security requires organizations to implement a comprehensive set of security and privacy controls to protect information systems and organizational operations. These controls include access control, identification and authentication, audit logging, incident response, risk assessment, and system monitoring. NIST emphasized that implementing these safeguards helps organizations preserve the confidentiality, integrity, and availability of information while reducing cybersecurity risks. For law firms, applying these controls strengthens the protection of confidential client records and supports compliance with legal and regulatory requirements.

The International Organization for Standardization (2022) stated that organizations should establish appropriate information security controls to protect information assets from unauthorized access, disclosure, modification, and destruction. ISO/IEC 27002 provides guidance on implementing controls related to access management, cryptography, physical security, asset management, logging and monitoring, incident management, and information protection. Applying these controls helps organizations strengthen their information security

management systems and improve resilience against information security risks. These recommendations are particularly valuable for law firms that regularly process confidential legal information.

C. Data Privacy

Data privacy refers to the protection of personal information and the right of individuals to control how their personal data is collected, used, processed, stored, and shared by organizations. It emphasizes that personal information should be handled in a lawful, fair, and transparent manner and only for legitimate purposes. Organizations are also expected to collect only the data that is necessary and to implement appropriate safeguards to prevent unauthorized access, disclosure, alteration, or misuse of personal information.

According to the Organization for Economic Co-operation and Development (OECD) (2013), organizations should follow internationally recognized privacy principles, including collection limitation, purpose specification, use limitation, security safeguards, openness, individual participation, and accountability. These principles provide guidance for responsible data processing and help organizations strengthen the protection of personal information. For law firms, maintaining strong data privacy practices is essential because they regularly collect and process highly sensitive personal information and confidential legal records. Protecting this information helps preserve client trust, uphold professional ethics, and support compliance with applicable data protection regulations.

The European Union (2016) explained that personal data must be processed lawfully, fairly, and transparently while ensuring that it is collected only for specified, explicit, and legitimate purposes. The General Data Protection Regulation (GDPR) also requires organizations to implement appropriate technical and organizational measures to protect personal data against unauthorized access, accidental loss, destruction, or disclosure. In

addition, organizations are expected to demonstrate accountability by complying with data protection principles throughout the data processing lifecycle. These principles are applicable to law firms because they regularly process confidential client information and must ensure that personal data is handled responsibly.

The National Privacy Commission (2022) emphasized that organizations processing personal information must adopt privacy management programs and implement appropriate organizational, physical, and technical security measures to protect personal data. The Commission explained that protecting personal information is essential to safeguarding individuals' privacy rights while promoting trust in digital services. Organizations are likewise encouraged to ensure that employees understand their responsibilities regarding data privacy and information protection. These practices are particularly important for law firms because they

The Organization for Economic Co-operation and Development (2013) explained that organizations should establish privacy policies based on internationally recognized principles such as collection limitation, purpose specification, use limitation, security safeguards, openness, individual participation, and accountability. The framework encourages organizations to implement responsible data governance practices that protect personal information while supporting the free flow of information across jurisdictions. Applying these principles helps organizations improve transparency, strengthen public trust, and ensure responsible data processing. For law firms, adherence to these principles contributes to the protection of sensitive client records and compliance with privacy regulations.

D. Data Privacy Act of 2012 (Republic Act No. 10173)

The Data Privacy Act of 2012 serves as the primary law governing the protection of personal information in the Philippines. Under Republic Act No. 10173 (2012), organizations are required to collect, process, store, and share personal information in a lawful and responsible

manner while protecting the privacy rights of individuals. The law also requires organizations to implement appropriate organizational, technical, and physical security measures to prevent unauthorized processing, accidental disclosure, or misuse of personal information.

Compliance with the law helps organizations avoid legal penalties and strengthens public confidence in how personal information is managed. For law firms, following the Data Privacy Act is important because they routinely process confidential client records and sensitive legal documents.

The National Privacy Commission (2022) explained that the Data Privacy Act of 2012 protects the fundamental human right to privacy while ensuring the free flow of information to promote innovation and economic growth. The Commission emphasized that organizations processing personal information must implement appropriate organizational, physical, and technical security measures to safeguard personal data against unauthorized access, accidental disclosure, alteration, destruction, and other unlawful processing. Compliance with these requirements enables organizations to strengthen data protection and maintain public trust. For law firms, these measures are essential because they regularly handle confidential client records and sensitive legal information.

Republic Act No. 10173 (2012) provides the legal framework governing the protection of personal information in the Philippines. The law requires organizations to process personal information lawfully, implement appropriate safeguards, respect the rights of data subjects, and ensure accountability in handling personal data. It also establishes the responsibilities of personal information controllers and processors, as well as penalties for violations involving unauthorized processing, negligent access, improper disposal, and unlawful disclosure of personal information. Compliance with these legal obligations is particularly important for law

firms because they process sensitive personal and legal information that requires a high level of protection.

E. Cybercrime Prevention Act of 2012 (Republic Act No. 10175)

The Cybercrime Prevention Act of 2012 provides the legal framework for addressing crimes committed through computers, networks, and other digital technologies. As mentioned by Republic Act No. 10175 (2012), the law covers offenses such as illegal access, computer-related fraud, identity theft, data interference, cyber libel, and other forms of cybercrime. It also supports the investigation and prosecution of cybercriminals while promoting cybersecurity in both the public and private sectors. Organizations are encouraged to strengthen their cybersecurity measures to reduce the risk of cybercrime and protect their digital resources. Law firms can benefit from understanding this law because it supports the protection of confidential legal information and digital evidence.

F. Information Security Governance

Information security governance helps organizations ensure that cybersecurity supports business goals and legal responsibilities. It involves leadership, policy development, risk management, accountability, and continuous improvement. As written in International Organization for Standardization (2020), information security governance enables organizations to manage security risks while ensuring that security activities remain aligned with organizational objectives. Strong governance also promotes better decision-making, improves compliance, and encourages management involvement in protecting organizational information. For law firms, effective governance helps ensure that security policies are consistently implemented and that confidential client information receives appropriate protection.

G. ISO/IEC 27001 Information Security Management System

ISO/IEC 27001 is one of the most widely recognized international standards for information security management. As stated by the International Organization for Standardization (2022), the standard provides organizations with a structured approach for identifying risks, implementing security controls, monitoring security performance, and continuously improving information security. Following ISO/IEC 27001 allows organizations to establish an Information Security Management System (ISMS) that helps protect sensitive information and demonstrate compliance with recognized security standards. Organizations that adopt this framework are generally better prepared to manage cyber risks and improve their overall cybersecurity posture.

H. Compliance Management

Compliance management refers to the process of ensuring that organizations follow applicable laws, regulations, policies, and industry standards. It includes identifying legal obligations, monitoring compliance activities, documenting evidence, and correcting compliance issues when necessary. As mentioned by the International Organization for Standardization (2021), an effective compliance management system promotes accountability, improves organizational governance, and reduces legal and operational risks. For organizations that manage sensitive information, such as law firms, compliance management helps maintain legal responsibilities while supporting secure and ethical business operations.

I. Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a security approach that limits access to information based on the responsibilities of individual users within an organization. Instead of giving every user the same level of access, permissions are assigned according to specific job

roles. This helps reduce unauthorized access, protects confidential information, and simplifies user management. Ferraiolo et al. (2016) explained that RBAC improves information security by ensuring that users can access only the information needed to perform their assigned duties. In law firms, implementing RBAC helps protect confidential legal documents by restricting access to authorized lawyers and staff members.

J. Audit Trail

An audit trail is a chronological record of activities performed within an information system. It records user actions such as logins, file access, data modifications, and administrative activities. Maintaining an audit trail helps organizations monitor system usage, investigate security incidents, identify unauthorized activities, and demonstrate compliance with legal and regulatory requirements. As found by Kent and Souppaya (2006), audit logs are an important component of information security because they support accountability, incident investigation, and continuous monitoring. For law firms, audit trails provide evidence of system activities and help protect confidential client information by ensuring transparency and accountability.

K. Legal Case Management System

A Legal Case Management System is a technology-based solution designed to help law firms and legal organizations organize, manage, and monitor case-related information efficiently. It provides a centralized platform for storing legal documents, tracking case progress, managing schedules, recording client information, and improving communication among legal professionals. According to Brooks (2020), legal technology solutions such as case management systems support law firms by improving document organization, automating routine tasks, and enhancing the management of legal information. Furthermore, integrating security features such

as access control, audit trails, and data protection mechanisms is essential because legal case management systems store confidential client information and sensitive legal documents.

For law firms, implementing a secure Legal Case Management System helps improve operational efficiency while ensuring that client records are protected. The integration of cybersecurity, data privacy, and compliance practices within the system supports responsible information management and helps organizations meet legal and ethical obligations in handling confidential legal information.

2.4 Synthesis of the Reviewed Literature and Studies

Based on the reviewed foreign and local studies, literature, and industry reports, cybersecurity and legal compliance are strongly interconnected in modern organizations.

Foreign studies emphasize that integrated cybersecurity-compliance systems improve security performance, reduce breaches, and strengthen organizational trust. Local studies highlight that Philippine institutions still struggle with manual compliance systems, lack of automation, and fragmented security tools.

Related literature supports structured frameworks such as NIST, ISO/IEC 27001, GDPR, and the Data Privacy Act of 2012.

However, these frameworks are often implemented separately rather than in unified systems, creating inefficiencies and security gaps.

Despite the availability of cybersecurity tools and compliance frameworks, there remains a clear gap in fully integrated platforms specifically designed for law firms. This gap supports the development of LEXSHIELD.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This research design focuses on creating, improving, and evaluating the LEXSHIELD system. It helps ensure that the platform meets the needs of lawyers, clients, and administrators while providing secure and reliable legal services.

This study will employ a Developmental Research Design because it focuses on the design, development, implementation, and evaluation of the proposed system, **LEXSHIELD: Integrated Cybersecurity and Legal Compliance Platform for Law Firms**. Developmental research is appropriate because the study aims to create a functional web-based platform that addresses cybersecurity risks while improving legal service management and regulatory compliance within law firms.

The study will begin by identifying the requirements of lawyers, clients, and administrators through interviews, surveys, document analysis, and a review of existing legal management systems. The collected information will serve as the basis for designing and

developing the system. After development, the system will undergo comprehensive testing and evaluation to determine whether it satisfies the functional, security, usability, and compliance requirements of its intended users.

3.2 System Development Methodology

Agile Software Development Methodology

The Agile methodology allows the researchers to develop LEXSHIELD in small stages. It also helps improve the system by making changes based on testing results and user feedback throughout the development process.

The Agile methodology consists of the following phases:

1. Requirements Analysis

- In this phase, the researchers will identify the needs and requirements of the proposed system. Information will be gathered from the target users through surveys, interviews, and related studies. The researchers will analyze the collected data to determine the features and functions that should be included in LEXSHIELD. One of the identified requirements is an email verification feature for newly registered clients to validate their accounts before they can access the system. The system requirements will serve as a guide for the design and development process. This phase helps make sure that the system meets the needs of its intended users.

2. System Design

- In this phase, the researchers will create the overall design of the proposed system. The system structure, user interface, database, and workflow will be planned based on the gathered requirements. The researchers will also prepare diagrams and layouts to show how the system will work. The researchers will also design an email verification process using SMTP through PHPMailer to verify newly created client accounts. This phase serves as a guide for the development of LEXSHIELD. A well-planned design helps make the system easier to develop and use.

3. Testing

- In this phase, the researchers will test the developed system to make sure that all features work properly. Different testing procedures will be conducted to check the system's functionality, security, and performance. The researchers will test whether only verified email accounts are allowed to access the system. Any errors or problems found during testing will be fixed by the researchers. The system will also be checked to ensure that it meets the requirements identified during the earlier phases. This phase helps ensure that LEXSHIELD is reliable and ready for evaluation.

4. Deployment (Prototype)

- In this phase, the completed prototype of LEXSHIELD will be prepared for presentation and evaluation. The researchers will install and demonstrate the system to the selected users so they can test its features and functions. Feedback and suggestions will be collected to identify areas that need improvement. Any necessary revisions will be made to enhance the prototype. This phase helps determine if the system is ready for its intended purpose.

3.3 System Architecture

This section explains how the different parts of LEXSHIELD work together to process data and provide services to users. The system will use a client-server architecture to separate the user interface, system functions, and database. This structure helps improve the system's security, performance, and maintenance. It also allows each layer to perform its specific role without affecting the others.

The three main layers of the system are:

1. Presentation Layer

- This is the part of the system that users interact with. It includes the web pages, forms, and dashboards where users can access the features of LEXSHIELD and perform different tasks.

2. Application Layer

- This layer handles the main functions of the system. It processes user requests, applies the system's logic, and manages features such as cybersecurity monitoring, compliance checking, verification and user authentication.

3. Database Layer

- This layer stores and manages all the data used by the system. It keeps user accounts, legal documents, compliance records, system logs, and other important information organized and secure.
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3.4 Tools and Technologies Used

This section explains the software and technologies that will be used to develop the LEXSHIELD system. These tools will help the researchers design, build, test, and manage the system throughout the development process. Each technology has a specific purpose in creating a functional and secure platform.

- **Frontend: HTML, CSS, JavaScript**

These technologies will be used to create the user interface of LEXSHIELD. They will help design the web pages, forms, dashboards, and other visual elements that users will interact with. They will also make the system responsive and easy to use.

- **Backend: PHP / Python / Node.js**

The backend technology will handle the main functions of the system. It will process user requests, manage system features, connect to the database, and ensure that the system works properly. It will also support the security and overall performance of LEXSHIELD.

- **Database: MySQL**

MySQL will be used to store and manage the data of the system. It will keep records such as user accounts, legal documents, compliance information, and other important information. This helps organize and protect the data stored in LEXSHIELD.

- **Development Tools: Visual Studio Code, Git, GitHub, XAMPP, Composer, phpMyAdmin, Google Chrome, Microsoft Edge, PHPMailer (SMTP)**

These development tools will be used throughout the creation of the LEXSHIELD system. They will help the researchers write, test, debug, and manage the source code efficiently. They will also support database management, version control, and team collaboration during development. This helps improve account security and prevents fake or invalid registrations. Using these tools will improve the quality and organization of the project. Together, they will help ensure that the system is developed successfully and functions as expected.

3.5 Data Gathering

Techniques This section explains the methods that will be used to collect the information needed to develop LEXSHIELD. The collected data will help the researchers understand the needs of the target users, identify cybersecurity concerns, and determine the features that should be included in the system. The information gathered will also serve as the basis for the design and development of the proposed platform.

The data gathering will be using according to these following ways:

- **Interviews**

Interviews will be conducted with lawyers, legal professionals, and IT experts to gather information about their experiences and needs. The researchers will ask questions about their current workflow, cybersecurity practices, and challenges they encounter. The information collected will help improve the design and functionality of LEXSHIELD.

- **Surveys and Questionnaires**

Surveys and questionnaires will be distributed to the target users to collect their opinions and suggestions. The researchers will gather information about the features they need, the problems they commonly experience, and their expectations for the proposed system. The responses will be used to support the development of LEXSHIELD.

- **Document Analysis**

Document analysis will be conducted by reviewing related laws, cybersecurity standards, research studies, and other references. The researchers will use these documents to understand legal requirements and cybersecurity practices that should be applied in the system. This will help ensure that LEXSHIELD follows relevant regulations and best practices.

- **Review of Existing Systems**

The researchers will review existing legal management and cybersecurity systems to understand how they work. Useful features, strengths, and weaknesses will be identified and analyzed. The information gathered will help improve the design of LEXSHIELD and avoid common problems found in similar systems.

3.6 Testing Procedures

This section explains the testing methods that will be used to evaluate the quality of LEXSHIELD. The researchers will conduct different tests to make sure that the system works properly, performs its intended functions, and provides a secure and reliable platform for users. The results of these tests will help identify and fix any problems before the system is evaluated.

- **Unit Testing**

Unit testing will be conducted to check whether each feature or module of LEXSHIELD works correctly on its own. The researchers will test each part of the system separately to identify and fix any errors before combining them with other modules.

- **System Testing**

System testing will be conducted to evaluate the complete LEXSHIELD platform. The researchers will check whether all features and functions work as expected and whether the system meets the identified requirements.

- **Security Testing**

Security testing will be performed to examine the security features of LEXSHIELD. The researchers will verify that user accounts, documents, and other sensitive information are properly protected against common security threats and unauthorized access.

- **User Acceptance Testing (UAT)**

User Acceptance Testing (UAT) will be conducted by allowing the intended users to use the system and evaluate its features. The researchers will collect their feedback on the system's

usability, functionality, and overall performance. The suggestions gathered during this phase will be used to improve the final version of LEXSHIELD.

3.7 Evaluation Criteria

This evaluation will determine whether LEXSHIELD meets its intended objectives and provides a functional, secure, and user-friendly platform for its intended users. The system will be evaluated based on the following criteria:

- **Functionality**

This criterion evaluates whether the features and functions of LEXSHIELD operate correctly and meet the system requirements. It ensures that users can perform their intended tasks successfully.

- **Usability**

This criterion measures how easy the system is to learn and use. It evaluates whether users can navigate the interface and complete tasks efficiently with minimal difficulty.

- **Security**

This criterion assesses how well LEXSHIELD protects user accounts, legal documents, and other sensitive information. It evaluates security features such as authentication, access control, and data protection.

- **Compliance Accuracy**

This criterion evaluates whether LEXSHIELD supports the legal compliance and cybersecurity requirements identified in the study. It verifies that the system follows applicable Philippine laws, standards, and project requirements.

- **Efficiency**

This criterion measures how quickly and effectively LEXSHIELD performs its functions. It evaluates the system's response time, processing speed, and overall performance while handling user requests.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

This diagram shows how data moves between the users, the LEXSHIELD system, and the database. It provides a simple overview of how information is entered, processed, stored, and returned by the system. It also helps guide the development of LEXSHIELD.

Data flow diagram of LEXSHIELD (legal services platform)

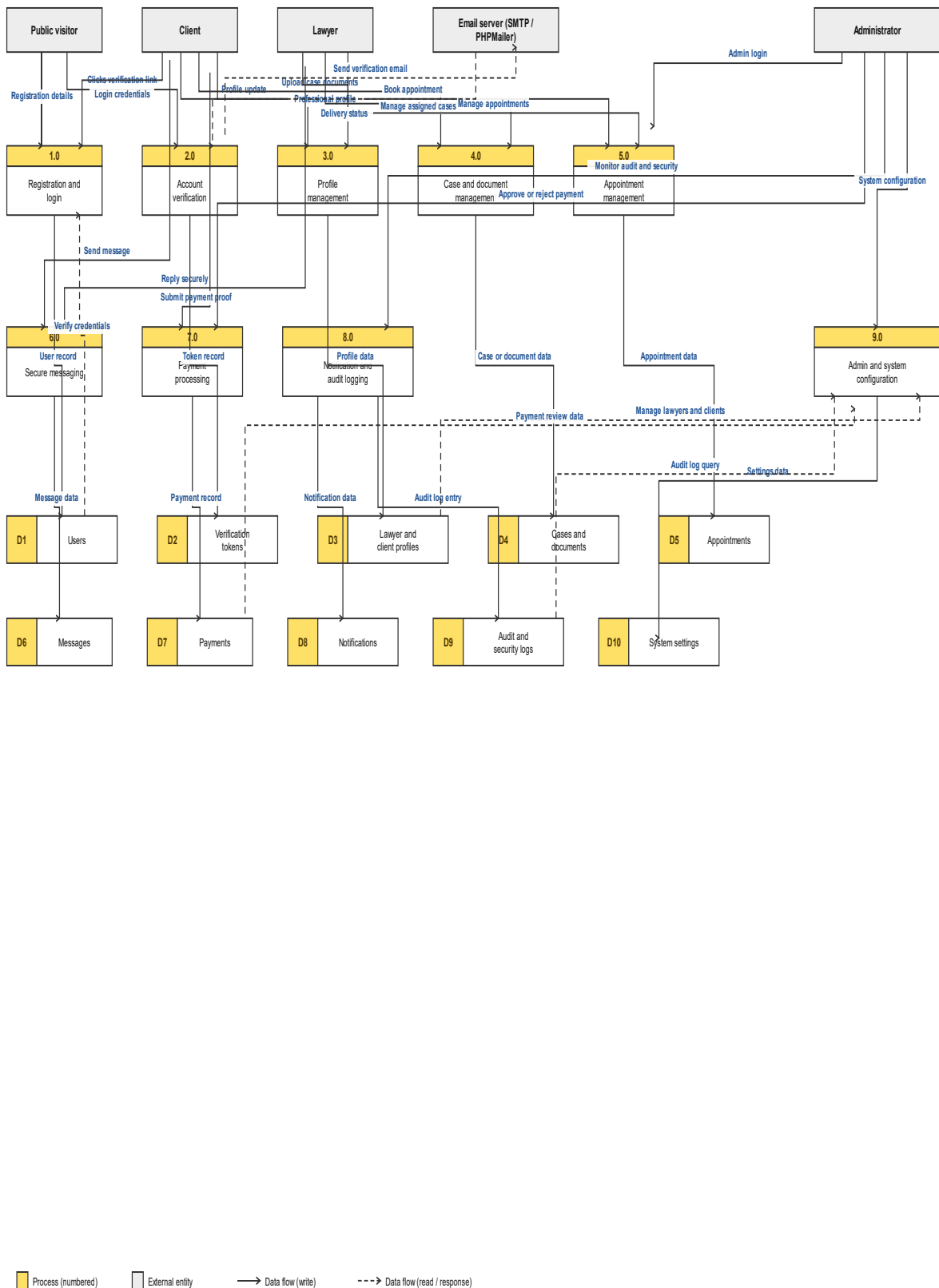
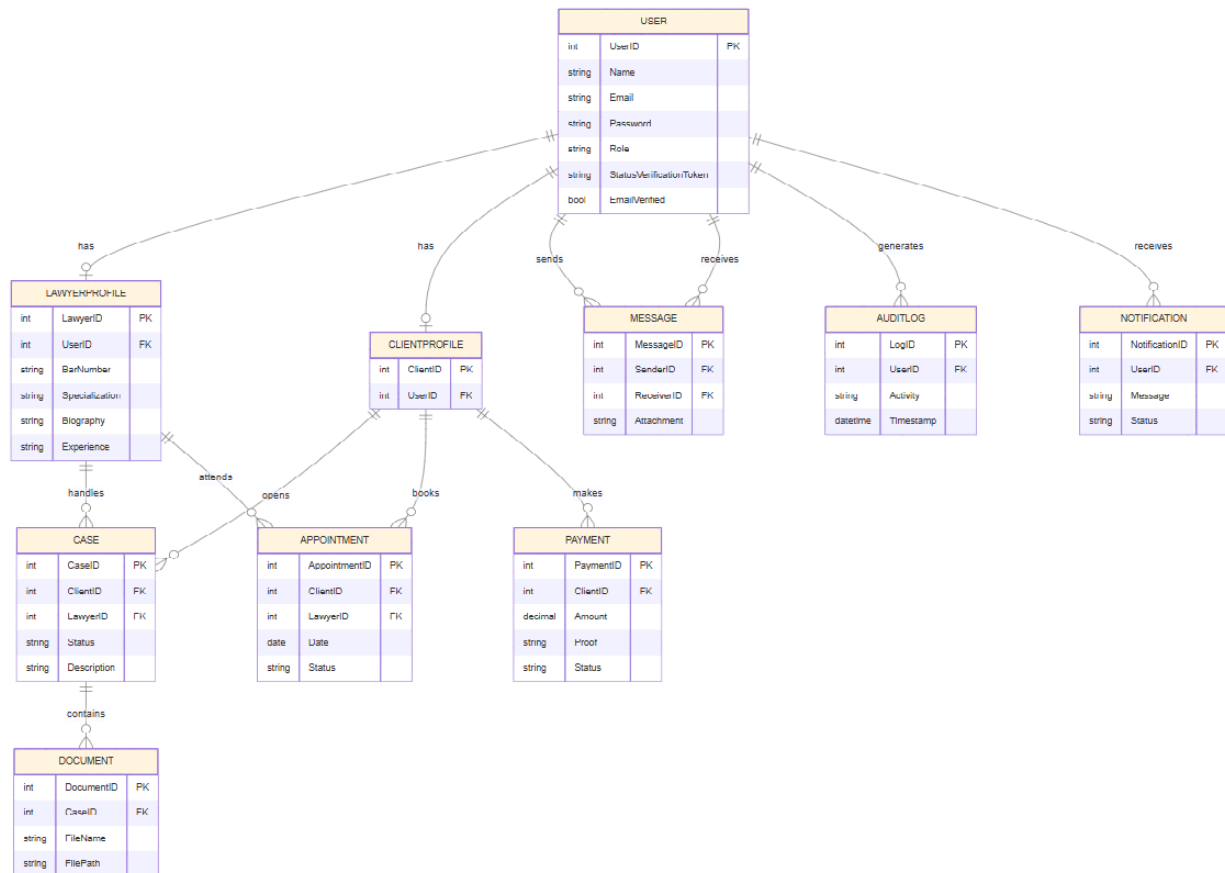


Figure - Data Flow Diagram of LEXSHIELD (User, Case & Account Management)

Entity Relationship Diagram (ERD)

The **Entity Relationship Diagram (ERD)** shows the database structure of LEXSHIELD. It illustrates the main entities, their attributes, and how they are related to each other. This helps the researchers organize and manage the system's data efficiently.

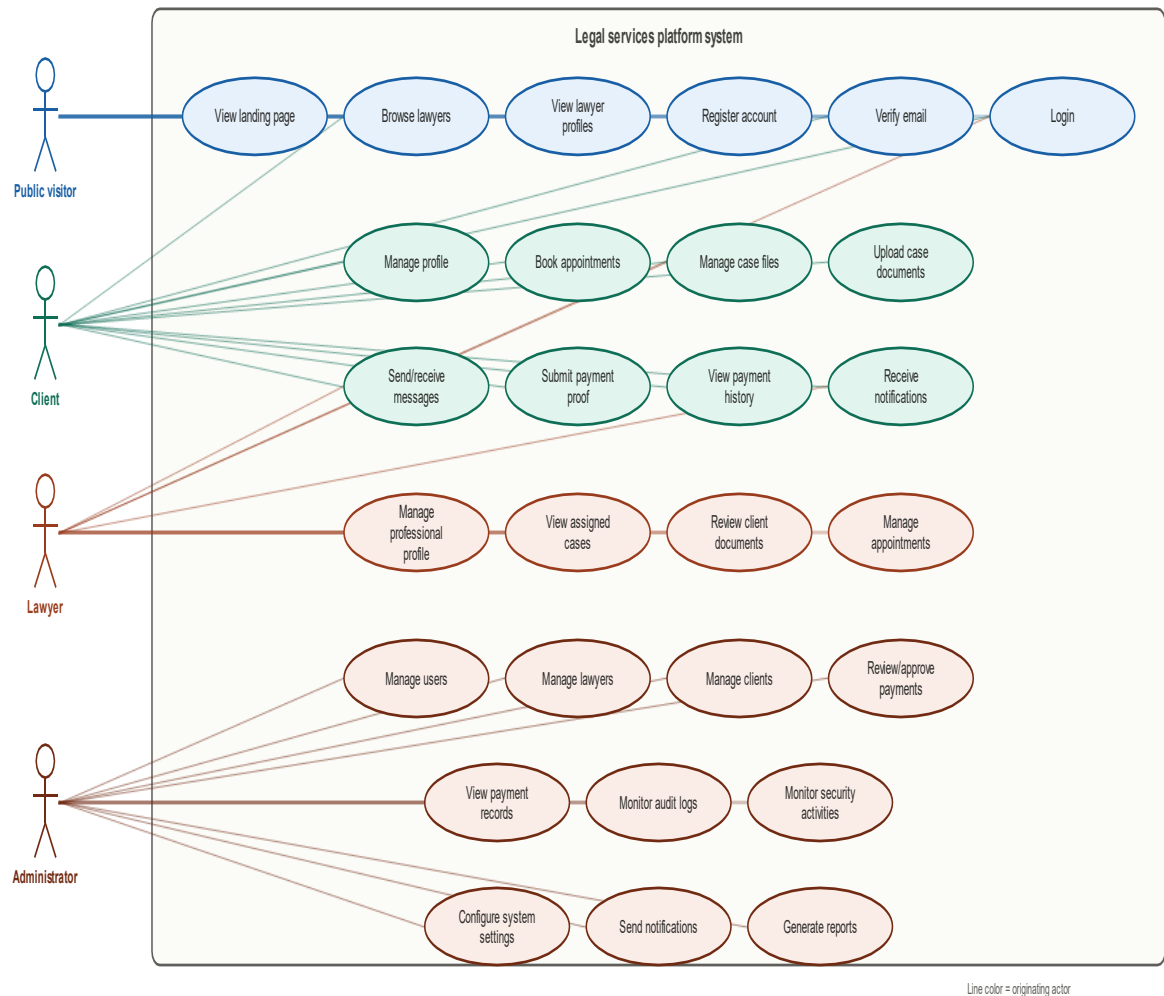
Entities:



Use Case Diagram

The **Use Case Diagram** shows the different users of LEXSHIELD and the actions they can perform within the system. It helps the researchers understand how users interact with the system and its main features

Legal services platform - use case diagram



End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.

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